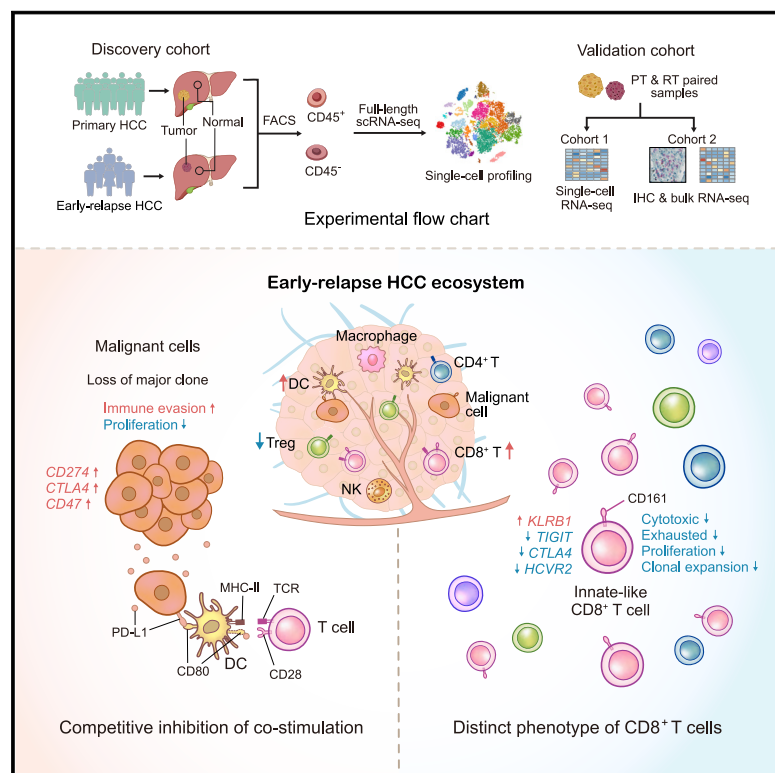


Single-cell landscape of the ecosystem in early-relapse hepatocellular carcinoma

Graphical Abstract



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In Brief

Single-cell analysis of primary and relapsed hepatocellular carcinoma tumors from patients reveal innate-like CD8⁺ T cells with low cytotoxicity and clonal expansion in the latter that may explain the compromised antitumor immunity and poor prognosis associated with liver cancer.

Highlights

- ScRNA-seq reveals a distinct immune ecosystem in early-relapse HCC
- Decreased Tregs, increased DCs, and CD8⁺ T cells were observed in early-relapse HCC
- CD8⁺ T cells have an innate-like, low cytotoxic, and low clonal expansion phenotype
- Recurrent malignant cells mediate the compromised antitumor immune response



Article

Single-cell landscape of the ecosystem in early-relapse hepatocellular carcinoma

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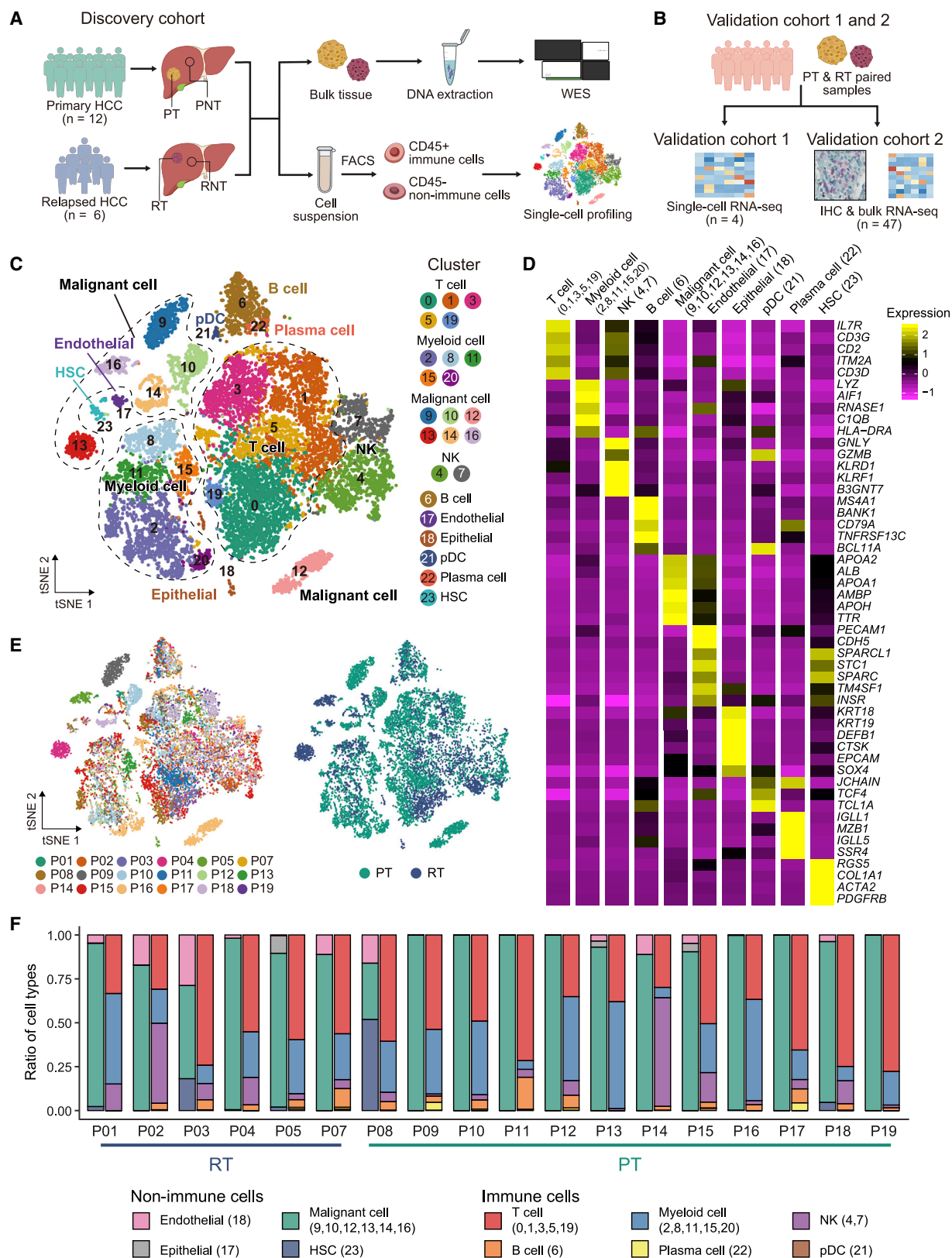
SUMMARY

Hepatocellular carcinoma (HCC) has high relapse and low 5-year survival rates. Single-cell profiling in relapsed HCC may aid in the design of effective anticancer therapies, including immunotherapies. We profiled the transcriptomes of ~17,000 cells from 18 primary or early-relapse HCC cases. Early-relapse tumors have reduced levels of regulatory T cells, increased dendritic cells (DCs), and increased infiltrated CD8⁺ T cells, compared with primary tumors, in two independent cohorts. Remarkably, CD8⁺ T cells in recurrent tumors overexpressed *KLRB1* (CD161) and displayed an innate-like low cytotoxic state, with low clonal expansion, unlike the classical exhausted state observed in primary HCC. The enrichment of these cells was associated with a worse prognosis. Differential gene expression and interaction analyses revealed potential immune evasion mechanisms in recurrent tumor cells that dampen DC antigen presentation and recruit innate-like CD8⁺ T cells. Our comprehensive picture of the HCC ecosystem provides deeper insights into immune evasion mechanisms associated with tumor relapse.

INTRODUCTION

Liver cancer is the fourth-highest cause of cancer-related deaths worldwide (Bray et al., 2018). Hepatocellular carcinoma (HCC) is the predominant form of liver cancer, accounting for 75%–85% of cases (Singal et al., 2020), for which surgical resection represents the most effective treatment, with curative potential (Zhou et al., 2020). However, the high incidence of tu-

mor relapse (50%–70% 5 years after surgery) has hindered improved survival (European Association for the Study of the Liver, 2018). Furthermore, early recurrence within 2 years after surgery accounts for 70% of relapsed HCC cases, is rarely treatable, and has been associated with poor survival (Zheng et al., 2017b). The molecular mechanisms associated with rapid postoperative recurrence remain poorly understood. A thorough exploration of tumor relapse could enhance our



(legend on next page)

understanding of the mechanisms associated with tumor development and progression and help discover more effective therapeutic strategies for HCC.

Tumors are complex ecosystems, defined by spatiotemporal interactions among heterogeneous cell types, including malignant, immune, and stromal cells (Wagner et al., 2019). Understanding the interplay of cell types is essential to understand tumor development, prognosis, and treatment. Recently, immune checkpoint blockades (ICBs) have shown clinical benefits in several types of cancer, although the response rates in HCC were unsatisfactory (El-Khoueiry et al., 2017; Finkelmeier et al., 2019; Kambhampati et al., 2019; Yau et al., 2019). The immune microenvironment might affect the treatment efficacy of ICBs (House et al., 2020). Therefore, dissecting the tumor ecosystem in HCC could provide insights into immune evasion mechanisms which might be helpful to design more effective immunotherapeutic strategies for HCC, especially for patients with relapsed tumors. Single-cell RNA sequencing (scRNA-seq) is a powerful tool for the investigation of the cellular components and their interactions in the tumor microenvironment (Azizi et al., 2018; Chevrier et al., 2017; Lambrechts et al., 2018; Lavin et al., 2017; Tirosh et al., 2016). ScRNA-seq has also been used to elucidate the association between some tumor microenvironmental patterns and clinical outcomes in several cancer types (Guo et al., 2018; Savas et al., 2018). Furthermore, several studies have profiled the single-cell landscape of immune cells and tumor cells' heterogeneity in primary HCC (Lim et al., 2019; Ma et al., 2019; Zhang et al., 2019; Zheng et al., 2017a). However, a comprehensive depiction of the ecosystem and immune phenotypes in early-relapse HCC at the single-cell resolution level remains lacking.

Here, we dissected the tumor immune microenvironment by examining 16,498 full-length single-cell transcriptomes from 18 primary or early-relapse HCC patients. We observed reduced levels of T-regulatory cells (Tregs) and increased fractions of dendritic cells (DCs) and CD8⁺ T cells in the immune component of early-relapse tumors (RT), compared with those in primary tumors (PT). Interestingly, CD8⁺ T cells in RT, characterized by overexpression of *KLRB1* (CD161), primarily presented an innate-like, low-cytotoxic, and low-clonal-expansion state, with the low expression of co-stimulatory and checkpoint molecules. Patients with more CD161⁺CD8⁺ T cells showed a worse prognosis. We validated these findings in two additional cohorts of paired PT and RT samples, using scRNA-seq or immunohistochemical (IHC) staining. Our analysis showed that malignant cells in RT have higher immune evasion characteristics, which potentially inhibit the capacity of DCs to activate CD8⁺ T cells.

Our data shed light on the unique aspects of the altered immune response associated with HCC relapse, which might guide the development of rational immunotherapies to benefit a wide range of patients.

RESULTS

Study design

To resolve the tumor ecosystem in early-relapse HCC, we collected surgical tumor specimens and paired adjacent non-tumor tissues from 12 treatment-naïve primary and 6 early-relapse HCC patients (discovery cohort) for scRNA-seq (Figure 1A). Tumors and paired adjacent non-tumor tissues from primary HCC are referred to as PT and primary non-tumor (PNT), respectively, whereas tumors and paired, adjacent, non-tumor tissues from early-relapse HCC are referred to as RT and relapsed non-tumor (RNT). Detailed clinical and pathological information, including tumor stage, alpha-fetoprotein (AFP) levels, and tumor size, are provided in Table S1. Moreover, we recruited four additional patients, with paired PT and early RT samples for scRNA-seq (validation cohort 1), and additional 47 patients, with paired PT and early RT samples to perform IHC and bulk RNA-sequencing validation (validation cohort 2). Validation cohorts were used to confirm differences found in the discovery cohort (Figure 1B).

In the discovery cohort, we performed bulk whole-exome sequencing (WES) of cell populations from primary HCC (P08-P19) and early-relapse HCC (P01-P05, P07) tissues and paired PT (P01-P05, P07) samples (Figure 1A; STAR methods). We confirmed that early-relapse HCCs that were investigated here represent recurrent, rather than *de novo* tumors. They shared more mutations (1,349–3,299; 27%–49% of total mutations in RT) with their paired PT samples than that shared by tumors of different origins from the same patients (0–12 mutations shared, 0%–0.28% of total mutations) (Yates et al., 2017). Phylogenetic analysis (Leshchiner et al., 2018), based on shared somatic single-nucleotide variants (SNVs) and insertions-deletions (INDELs), also show that the RT and paired PT samples are clonally related (Figure S1A).

Due to differences in patients' immune infiltration levels, we applied fluorescence-activated cell sorting (FACS) to isolate CD45⁺ and CD45[−] cells (Figure S1B). To obtain sufficient cell numbers to investigate the immune components in tumors, we sorted CD45⁺ and CD45[−] cells, at a ratio of 3:1 (1,500 versus 500 cells), from each patient, and generated scRNA-seq libraries, using the microwell full-length mRNA amplification and library construction system (MIRALCS) (Wu et al., 2015). A total of 16,498 cells passing the quality control stage were used for further analysis (119–1,617 cells; median: 926 cells / patient),

Figure 1. ScRNA-seq profiling of the primary and relapsed HCC tumor environments

(A and B) Schematic representation of the experimental strategy. The discovery cohort is shown in (A), and the validation cohorts 1 and 2 in (B). The numbers (n) of patients in each subgroup are provided in the figure.

(C) T-distributed stochastic neighbor embedding (t-SNE) plot, showing the annotation and color codes for cell types in the HCC ecosystem.

(D) Heatmap showing the expression of marker genes in the indicated cell types. The top bars label the clusters corresponding to specific cell types, and the number in brackets correspond to the cluster number in (C).

(E) The t-SNE plot, showing cell origins by color, patient origin (left panel), and PT or RT origin (right panel).

(F) Histogram indicating the proportion of cells in tumor tissue of each analyzed patient. Non-immune cells and immune cells are shown in separate histograms. See also Figure S1 and Tables S1, S2, and S7.

which generated ~12 million total mapped reads and 3,490 (transcripts per million [TPM] >1) detected genes, on average (Figure S1C; Table S2).

scRNA-seq profiling of the tumor ecosystem in primary and early-relapse HCC

To construct a global tumor niche atlas, we performed cell classification and marker gene identification using Seurat (STAR methods). We identified and visualized 24 clusters using the T-distributed stochastic neighbor embedding (t-SNE) method (Figure 1C; STAR methods). All cells (both immune and non-immune fractions) expressed high levels of housekeeping genes, such as *ACTB*, *B2M*, and *GAPDH* (Figure S1D), whereas immune cells were *PTPRC*⁺, as expected (Figure S1E), indicating the accuracy of our data.

Non-immune cells primarily consisted of endothelial cells (ECs; *PECAM1* and *CDH5*), hepatic stellate cells (HSCs; *ACTA2* and *PDGFRB*), apparently normal epithelial cells (*EPCAM* and *KRT19*), and HCC malignant cells (Figures 1D and S1E). Low proportions of ECs and HSCs (<5% of total non-immune cells) were observed in most patients (Figures 1E and 1F). We observed the enrichment of *THY1*, a hepatic cancer stem cell (CSC) marker (Yamashita et al., 2013), in HSCs (Figure S1F), suggesting that this gene may not be an accurate CSC marker. Our data also indicated that most non-immune cells in tumors were malignant cells, expressing hepatocyte-like genes (*ALB*, *TTR*, and *APOA2*) (Figures 1D and S1E). The tumor origins were confirmed by the detection of copy number variants (CNVs), inferred by scRNA-seq data (Figure S1G; STAR methods). Similarly, we noticed that cluster 12 (only observed in patient P16) showed low expression of hepatic genes and *AFP* (Figure S1E), consistent with the IHC results (Figure S1H). These cells were also defined as HCC malignant cells by inferred CNVs and covering SNVs detected at the bulk level (Figures S1G and S1I). In agreement with previous reports (Puram et al., 2017), malignant cells were separated by patients (Figures 1C and 1E). We did not observe clusters of normal hepatocytes, likely due to the well-known technical limitations associated with obtaining these cells from non-tumor liver tissues.

The identified immune cells included myeloid-derived cells (*LYZ* and *C1QB*), T cells (*CD3D/G*), B cells (*MS4A1* and *CD79A*), plasma cells (*IGLL1* and *MZB1*), and natural killer (NK) cells (*KLRF1* and *KLRD1*) (Figures 1C, 1D, and S1E). All these cell subtypes were shared among patients and between PT and RT samples, albeit at different proportions (Figures 1E and 1F). The infiltration levels of B cells, plasmacytoid DCs (pDCs), and plasma cells were relatively low in all patients. Other immune cell clusters varied among patients, revealing substantial heterogeneity of immune cell compositions among HCC tumors (Figure 1F). In summary, despite this variability, RT and PT samples shared the same major immune cell subtypes and similar proportions of lymphoid- and myeloid-derived cells (Figure S1J).

A distinctive immune ecosystem in early-relapse HCC

We next performed unsupervised clustering of myeloid cells and T cells/NKs. A total of 11 clusters emerged within the myeloid lineage, including five clusters for macrophages

(Macro1–Macro5), two for monocytes (Mono1 and Mono2), three for DCs (DC1–DC3), and one for actively cycling cells (Figures 2A–2C).

Macrophages were identified by high expression levels of *CD163* and *CD68*. The clusters Macro2–Macro5 showed patient bias, consistent with previous reports (Kumar et al., 2018), suggesting that the distinct microenvironment of each tumor contributes to the heterogeneity of this population (Figures 2B and S2A). Macrophages were significantly enriched in tumors compared with paired adjacent non-tumor tissues and displayed similar proportions in the immune fractions of RT and PT samples (Figures S2B and S2D). We could not clearly distinguish M1 and M2 macrophages using known marker genes such as *FCGR3A* (M1) and *CD163* (M2), as they were both expressed in those cells (Figure S2A), consistent with a previous report (Azizi et al., 2018). However, by calculating M1 and M2 polarization and pro- and anti-inflammatory scores using related gene sets (Table S3; STAR methods), we observed a dominant M2-like phenotype (Figure S2C). Mono1 cells displayed high expression levels of *FCN1*, *APOBEC3A*, and *THBS1* (Figures 2C and S2A), like the M-c1-THBS1 cells (Zhang et al., 2019) that were recently reported in HCCs. The Mono2 cluster was characterized by *PRF1*, *GNLY*, and *KLRC4-KLRK1* expression (Figure S2A), showing similar characteristics with reported Mono4 cells (Villani et al., 2017). None of these subtypes was associated with liver-resident Kupffer cells (MARCO⁺) (MacParland et al., 2018) or myeloid-derived suppressor cells (MDSCs, *CD33*⁺) (Eruslanov et al., 2012; Poschke and Kiessling, 2012) (Figure S2A).

Total DCs showed variation among patients, ranging from 0.1%–20% of the total immune fraction. This proportion was higher in RT than in PT samples and higher in RNT than in PNT samples (Figures 2D and S2D). Dual-color IHC staining verified the increased abundance of DCs (IDO⁺CD45⁺ or CD1c⁺CD45⁺) in RT versus PT from the validation cohort 2 (Figures 2E and 2F). HCC infiltrated DCs (HCC DCs) had unique transcriptional profiles compared with the reported datasets (Villani et al., 2017) for DCs in peripheral blood mononuclear cells (PBMCs). HCC DCs exhibited increased expression of major histocompatibility complex (MHC)-II molecules (*HLA-DQB2* and *HLA-DPB1*) and *BIRC3* and downregulation of MHC-I genes (*HLA-A* and *HLA-C*) compared with PBMC DCs (Figure S2E), indicating that MHC-II antigen-presentation was dominant in tumors. In contrast with PBMC DCs, only a portion of HCC DC1 cells (*CLEC9A*) expressed the classical DC1 marker gene *THBD* (CD141) (Figure S2F). DC2 cells (*CD1C* and *CLEC10A*), like previously reported CD1c⁺ DCs (Villani et al., 2017), also had unique signatures (Figure S2E). HCC DC2 cells showed a high expression of *CD1E* while lacking *CD14*, *VCAN*, *S100A9*, *SELL*, and *S100A8*, compared with PBMC DC2 cells (Figure S2F). We noticed that *AXL* was extremely enriched in HCC DC2 cells (Figure S2F), suggesting their migratory features. DC3 cells (*CCL19*, *LAMP3*, and *CCR7*) could represent a tumor-specific DC subpopulation, because they were also detected in our reanalysis of melanoma (Tirosh et al., 2016) (Figure S2G) and head and neck head cancer (Puram et al., 2017) datasets (Figure S2H), but were not found in PBMCs. According to published studies (Saban, 2014; Worbs et al., 2017; Zhang et al., 2019), DC3 cells have the potential to migrate to lymph nodes. Among the three

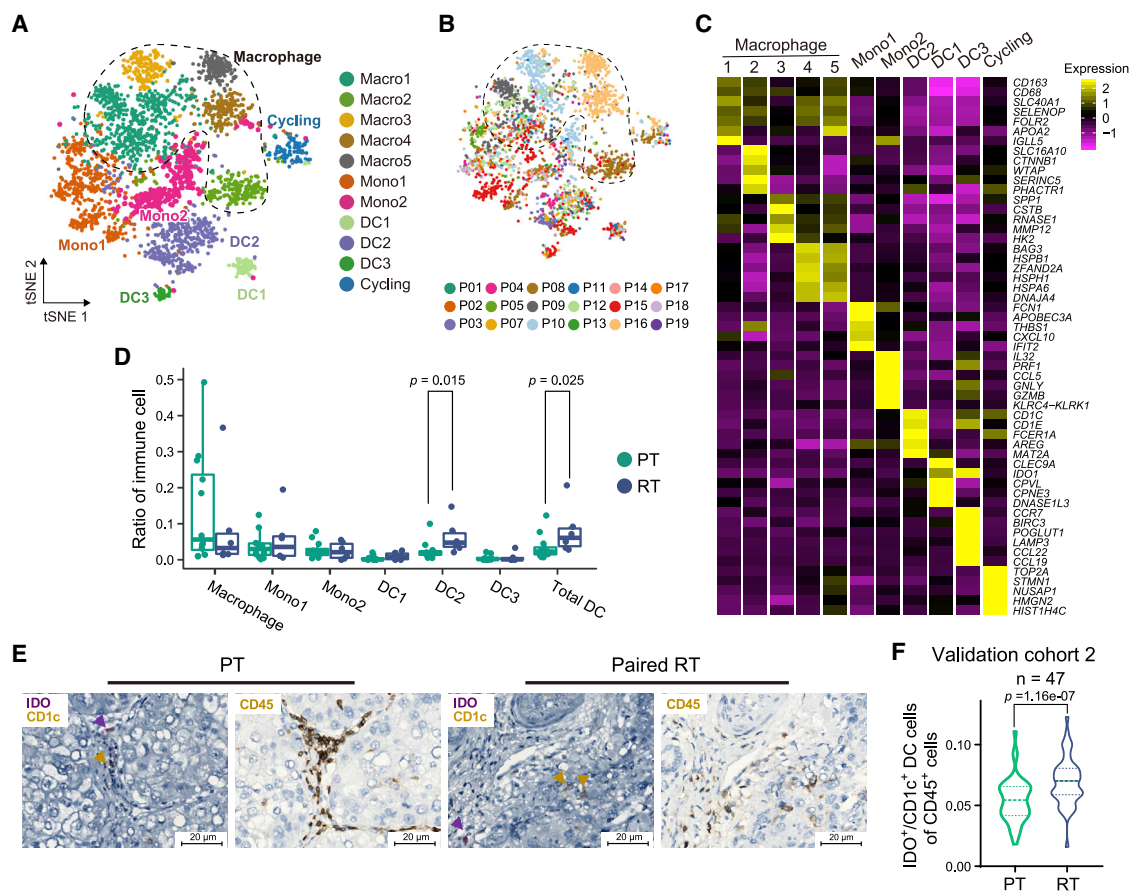


Figure 2. Myeloid-derived cell components in primary and relapsed HCC

(A) t-SNE plot showing the subtypes of myeloid-derived cells derived from PT and RT patients. Each cluster is color-coded according to cell type. Cluster annotations are indicated in the figure.

(B) t-SNE plot showing PT and RT myeloid cells, clustered and color-coded, according to patient.

(C) Heatmap showing the expression of marker genes in each subtype of myeloid-derived cells.

(D) Boxplot showing the fraction of myeloid subgroups in PT (green) and RT (blue).

(E) Representative images of IHC staining in Formalin-fixed paraffin-embedded (FFPE) tissues, indicating CD45⁺IDO⁺ (DC1 or DC3) or CD45⁺CD1c⁺ (DC2) cells, in paired RT and PT samples. Scale bar, 20 μ m.

(F) Violin plot, presenting the fractions of DCs in paired PT and RT samples from validation cohort 2, based on IHC staining results.

All statistical analyses are paired (F) or unpaired Wilcoxon tests (D). N represents the number of patients.

See also Figure S2 and Table S3.

DC subsets, only DC2, which presents specific antigens to CD4⁺ but not CD8⁺ T cells, resided in RT samples (Figure 2D).

The re-clustering of T cells/NKs revealed 17 populations, including two NK subtypes (NK CD16 and CD160), four subtypes of CD4⁺ T cells (CD4 CCR7, CD4 IL2, Treg FOXP3, and Treg LAYN), and seven clusters of CD8⁺ T cells (CD8 CCR6, CD8 NR4A1, CD8 GZMK, CD8 XCL1, CD8 CTLA4, and CD8 LAG3) (Figures 3A and S3A). All these subtypes were shared across patients and between PT and RT samples (Figure 3B). NK CD160 cells were enriched in adjacent non-tumor tissues (PNT and RNT) and showed high expression levels of *CD69*, *NR4A1*, *CD160*, and *CXCR6* and low level of *FCGR3A*, suggesting that they represent tissue-resident memory NK cells (Figure S3B). NK CD16 (*FCGR3A*) cells overexpressed cytotoxic genes, including *GNLY*, *GZMB*, and *PRF1*, the inhibitory-related

gene *HAVCR2* (Al Omar et al., 2011) (Figure 3C), and inhibitory KIRs (*KIR2DL-1* and *KIR2DL-3*) (Figure S3B). The high expression of *FCGR3A* suggested that these cells may be recruited from the peripheral blood (MacParland et al., 2018). NK CD16 cells were the primary NKs found in tumors, demonstrating that peripheral NKs are the primary source of tumor-infiltrating NKs (Figure S3D).

A higher diversity of T cells was observed in tumors than in adjacent non-tumor tissues (Figure S3E). Tregs (Treg FOXP3 and Treg LAYN) highly expressed *IL2RA*, *FOXP3*, and *CTLA4* (Figures 2I and S3A). Treg FOXP3 cells were enriched in tumors, particularly in PT samples (Figures 3D and S3C), and displayed increased expression of *CD27*, *FOXP3*, and *LAI2* (Figure S3A). Conversely, Treg LAYN cells showed increased expression of *LAYN*, *TNFRSF9*, and *ICOS* (Figure 3C), like the previously



that the transcriptional profile of CD8 GZMH cells was like the reported CD8 CXCR3 cells (Zheng et al., 2017a). Because CD8 NR4A1 cells showed the highest expression of activator protein 1 (AP-1; *JUN* and *FOSB*), *NR4A1*, *CD69*, and *RUNX3*, we designated them as liver-tissue-resident, memory-like T cells (Figures 3C and S3A). CD8 XCL1 cells highly expressed genes associated with cytotoxicity (*GZMA*, *GZMB*, and *GZMK*) and moderately expressed exhaustion-related markers (*TIGIT*, *CTLA4*, and *LAG3*) (Figure 3I and S3A). CD8 CTLA4 and CD8 LAG3 cells showed elevated expression of T cell exhaustion markers (*PDCD1*, *HAVCR2*, *CTLA4*, *TIGIT*, and *LAG3*) (Figures 3C and S3A). CD8 CCR6 and CD8 NR4A1 cells had a rising trend of proportions in RT than PT samples, whereas CD8 XCL1 cells were enriched in PT (Figure 3D). Although none of the CD8⁺ T cell subtypes showed significant differences between RT and PT samples (Figure 3D), the total fraction of CD8⁺ T subtypes were more abundant in RT versus PT (79.7% versus 39.9% of total T cells, $p < 0.001$; Figure 3E). In contrast, the proportions of CD4⁺ T cells displayed the opposite trend (Figure 3E).

In summary, relapsed HCC presents a unique immune ecosystem, with increased proportions of DCs, reduced Tregs, reduced T cell proliferation, and a higher abundance of CD8⁺ T cells compared with the primary HCC microenvironment (Figures 2D, 3C, and 3E). These observations were validated by performing scRNA-seq in the 4 PT and RT pairs from validation cohort 1 (Figure S3I) and by performing IHC in the 47 paired PT and RT samples from validation cohort 2 (Figures 2F, 3F, and S3J).

CD8⁺ T cells share the same transition trajectory but reside in different states in primary and early-relapse HCC

We next explored the dynamic immune states and cell transitions in HCC-infiltrated CD8⁺ T cells by inferring the state trajectories using Monocle (Figure 4A; STAR methods). In pseudotime analysis, we removed MAIT cells, due to their differing TCR characteristics, and CD8 LAG3 cells, due to low expression of *CD3D*/*G/E*. This analysis showed that the CD8 CCR6 cells were at the beginning of the trajectory path, whereas the CD8 CTLA4 cells and most CD8 XCL1 cells at a terminal state (Figure 4A). By integrating TCR clonality (Figure S4A; STAR methods) and trajectory information, this transition was determined to initiate with CD8 CCR6 and CD8 NR4A1 cells, through an intermediate cytotoxic state characterized by CD8 GZMH and CD8 GZMK cells, and finally reach an exhausted state, characterized by CD8 CTLA4 and CD8 XCL1 cells. Based on the TCR clonal expansion observed among various cell types, we observed that CD8 XCL1 cells shared more TCR clones with cytotoxic subtypes (CD8 GZMH and CD8 GZMK cells) than with CD8 CTLA4 cells (Figure S4A). Additionally, CD8 XCL1 cells expressed lower exhausted scores than CD8 CTLA4 cells (Figure S4B), revealing that these cells represent an earlier exhausted state. We confirmed that the exhausted signature was upregulated, whereas the cytotoxic signature, despite an initial increase, was downregulated during these transitions (Figure 4B; STAR methods).

To further delineate the transition states associated with CD8⁺ T cells in RT samples, we analyzed the trajectories of CD8⁺

T cells in PT and RT samples separately. Surprisingly, early-stage CD8⁺ T cells were predominantly distributed in RT samples, with few cells identified at the end of the cell state transition path, whereas CD8⁺ T cells in PT samples were primarily at the terminal ends of the cytotoxic and exhausted states (Figure 4C). We next investigated the transcriptional changes associated with transitional states and observed that the CD8⁺ T cell clusters could be categorized into 4 phases (Figure 4D). CD8 CCR6 and CD8 NR4A1 cells were predominantly phase 1 cells, characterized by upregulated expression of *CCR6*, *CCR7*, *NCR3*, and *KLRB1* and low expression of *GZMA*, *GZMB*, and *GZMH*, suggesting that these cells had the lowest cytotoxic capacities (Figures 4D and S4C). Pathway analysis indicated that signaling pathways involved in the metabolism of lipids and lipoproteins, and transforming growth factor (TGF)- β receptor signaling, were enriched in phase 1 (Figure 4D). The analysis of energy metabolism pathways showed that phase 1 cells exhibited the highest expression scores for lipogenesis, glycolysis, and glutamine metabolism and equivalent scores for the tricarboxylic acid (TCA) cycle and lipolysis (Figures 4E and S4D; STAR methods). Metabolic analysis suggested that the functions of phase 1 cells rely on fatty acid oxidation and aerobic glycolysis, rather than oxidative phosphorylation (OXPHOS). This finding indicated that phase 1 cells were different from classical naive cells (Zhang and Romero, 2018) and were more closely related to a memory phenotype (O'Sullivan et al., 2014). Alternative splicing analysis of *CD45* demonstrated that these cells prefer CD45RO rather than CD45RA, indicating that they have been previously activated rather than being naive. The analysis of transcription factors (TFs) expressed in these cells revealed significant upregulation of *FOXP1*, *KLF10*, *DNAJC2*, *BCL11B*, *ELK3*, and *NFKB1* (Figures 4F and S4E). *FOXP1* inhibits both cell proliferation and *GZMB* expression in CD8⁺ T cells (Wei et al., 2016), appearing as a hallmark of unresponsiveness, dampening CD8⁺ T cells antitumor effector functions. *FOXP1* also mediated TGF- β -induced c-JUN transcriptional repression, showing unique functions for survival maintenance in memory T cells (Stephen et al., 2014). *KLF10* plays a role in suppressing CD8⁺ T cell activation, as *Klf10*-deficient mice display more effector CD8⁺ T cells (Papadakis et al., 2015). Cells in phase 2 demonstrated the highest expression levels of *JUN*, *FOS*, *NR4A1*, and *NR4A3*, matching the phenotypes of liver resident memory cells (Figures 4D, 4F, and S4E). Phase 3 was characterized by increased expression levels of classical cytotoxic genes (*GZMH*, *GZMB*, *GZMA*, and *GNLY*) and low expression of T cell exhaustion markers (*PDCD1*, *CTLA4*, and *HAVCR2*) (Figures 4D, 4G, S4C, and S4E). Pathway analysis suggested that cells in phase 3 were dependent on fatty acid metabolism, consistent with effector phenotypes and fatty acids representing a primary energy resource for effector T cells. Phase 4 was characterized by high levels of T cell exhaustion-related TFs, including *EOMES*, *TOX*, *TOX2*, *HOPX*, *ARNT*, *ETV1*, and *IRF8*, and genes involved in programmed cell death protein 1 (PD-1) and mitotic pathways, further confirming the exhausted state of these cells (Figures 4D–4F and S4E).

Interestingly, the checkpoint molecules *PDCD1* and *LAYN* were only modestly upregulated during the transition between the cytotoxic (phase 3) and exhausted states (phase 4), whereas

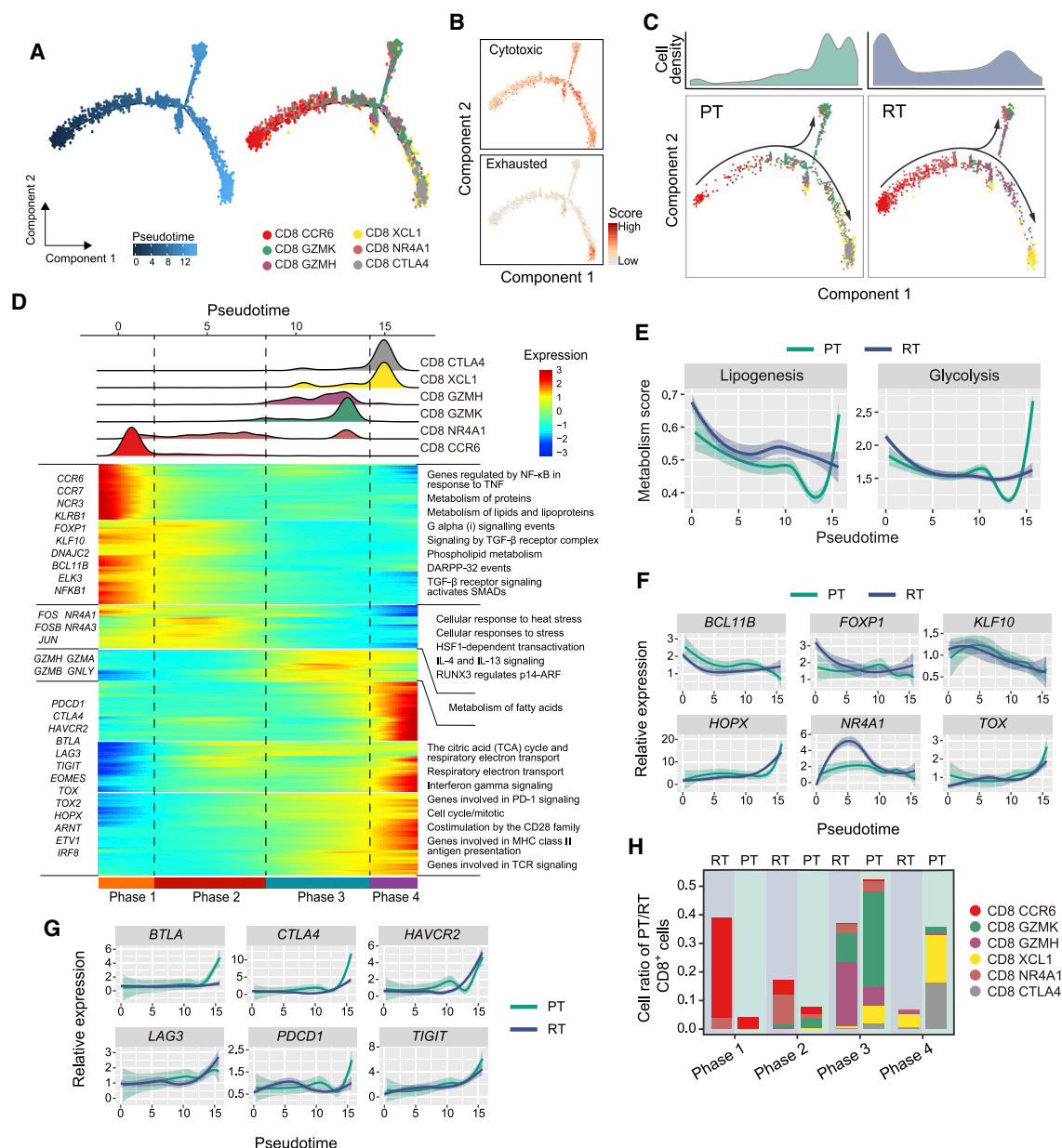


Figure 4. Analysis of CD8⁺ T cell transition states in RT and PT samples

(A) Pseudotime-ordered analysis of CD8⁺ T cells from PT and RT samples. T cell subtypes are labeled by colors.

(B) 2D pseudotime plot showing the dynamics of cytotoxic (upper panel) or exhausted signals (lower panel) in CD8⁺ T cells, from PT and RT samples.

(C) 2D graph of the pseudotime-ordered CD8⁺ T cells, from PT (left panel) and RT (right panel) samples. The cell density distribution, by state, is shown at the top of the figure.

(D) Heatmap showing the dynamic changes in gene expression along the pseudotime (lower panel). The distribution of CD8 subtypes during the transition (divided into 4 phases), along with the pseudo-time. Subtypes are labeled by colors (upper panel).

(E) Two-dimensional plots showing the expression scores for genes related to lipogenesis and glycolysis, in PT (green) and RT (blue) samples, along with the pseudotime.

(F and G) Two-dimensional plots showing the dynamic expression of transcription factors (F) and checkpoint genes (G) during the T cell transitions along the pseudotime.

(H) Histogram showing the cell distribution of CD8⁺ T cells, in RT and PT samples. CD8 subtypes labeled by colors.

See also Figure S4.

CTLA4 and *HAVCR2* expression increased by as much as 10-fold (Figures 4G and S4E). The elevated expression of *BTLA* was only observed in PT, whereas *LAG3* and *TIGIT* expression gradually increased from the inactivated to exhausted states (Figure 4G). These results demonstrated that the expression of immune checkpoints is heterogeneous during the CD8⁺ T cell state transition process (Figures 4G and S4E). Specific targeting strategies may be beneficial for the application of ICB therapeutic regimens to HCC patients. CD8⁺ T cells from RT samples were primarily characterized as phases 1 and 2, with only a few representing the exhausted phase 4, indicating a resident-memory phenotype. In contrast, CD8⁺ T cells in PT were predominantly characterized as phases 3 and 4, representing the transition process from cytotoxic to exhausted states (Figure 4H). Despite RT and PT samples displaying comparable fractions of phase 3 cells, they fell into different cytotoxic cell clusters, with RT samples associated with CD8 GZMH cells and PT samples associated with CD8 GZMK cells (Figure 4H).

We conclude that CD8⁺ T cells in PT and RT samples showed the same transition trajectories but displayed considerably different immune and transcriptional states, suggesting that different immune therapy strategies should be considered for the treatment of primary and recurrent HCC.

CD8⁺ T cells in early-relapse HCC reside in an innate-like phenotype

To further investigate the unique transcriptional states of CD8⁺ T cells in RT samples, we determined the expression scores of tissue-resident, cytotoxic, co-stimulatory, and exhausted T cell phenotypes (STAR methods). CD8⁺ T cells in RT samples showed reduced expression of exhausted signatures and a decreased fraction of cells expressing checkpoint genes, compared with CD8⁺ T cells in PT samples (Figures S5A and S5B). Tissue-resident genes, such as *RUNX3*, *NR4A1*, and *CD69*, increased, whereas co-stimulatory and exhausted molecules, such as *TNFRSF9*, *CD28*, *ICOS*, *TIGIT*, *CTLA4*, and *HAVCR2* were reduced (Figure 5A). Sc/single nucleus (sn)RNA-seq data for the four additional PT and RT paired samples in the validation cohort 1 also demonstrated increased tissue-resident gene levels and decreased inhibitory/cytotoxic signals in CD8⁺ T cells from RT samples (Figure S5C).

Differentially expressed gene (DEG) analysis indicated that the top downregulated genes in CD8⁺ T cells in RT included *CD27*, *TIGIT*, *CTLA4*, and *BTLA*, suggesting low inhibitory signals (Figure 5B; Table S4). Ingenuity pathway analysis (IPA) showed that CD8⁺ T cells in RT were characterized by downregulated OX40 and exhausted T cell signaling pathways, indicating that these cells are in a less activated but not exhausted state (Figure 5C). IPA also identified *PDGF-BB* as the top upstream factor that mediated this phenotype. Because *PDGF-BB* can inhibit T cell proliferation (Chen et al., 2014), we speculated that this might be a primary driver of the non-activated CD8⁺ T cell state in relapse HCC. Compared with cells in PT samples, CD8⁺ T cells in RT samples showed an enhanced response to oxidative stress, as indicated by the increased expression of *SOD1*, *SOD2*, *GADD45B*, *FOS*, *JUNB*, and *NFKB1* (Figures 5B and 5C; Table S4). However, those cells displayed downregulated *DUSP4* expression, suggesting an uncontrolled oxidative stress

response process (Dougherty et al., 2017). Also, these cells also strongly downregulated the genes in the oxidative phosphorylation pathway, suggesting possible mitochondrial dysfunction. Metabolic disorders might also contribute to CD8⁺ T cell dysfunction in RT.

DEG analysis also showed that CD8⁺ T cells in RT upregulated *KLRB1* (CD161) and *IL7R*, demonstrating an innate-like and memory T cell phenotype (Ruiz et al., 2014) (Figure 5B). The increased expression of *KLRG1*, *NCR3*, and *RORA* was also indicative of a type-2 innate lymphoid cell (ILC2)-like phenotype (Chiossone et al., 2018) (Figure 5B; Table S4). By performing dual-color IHC staining assay in the validation cohort 2, we verified the enrichment of CD8⁺CD161⁺ T cells in RT samples (median: 33.3% versus 16.7%, Figures 5D and 5E). To further assess the effector functions of these cells, we isolated tumor-infiltrated CD8⁺CD161⁺ and CD8⁺CD161[−] T cells and examined granzyme B (GZMB) expression levels by flow cytometry. CD8⁺CD161⁺ T cells displayed lower GZMB abundance than CD8⁺CD161[−] T cells, consistent with weaker cytotoxicity (Figures 5F and 5G). Furthermore, we found that the enrichment of CD161⁺CD8⁺ T cells in RT was significantly associated with a higher second recurrence rate (Figure 5H; $n = 47$, $p < 0.05$). Moreover, the comparison of transcriptional profiles between *KLRB1*^{high} CD8 T cells and *KLRB1*^{low} CD8 T cells identified 255 significant DEGs (Table S5). In *KLRB1*^{high} CD8 T cells, we identified significantly increased expression of *IL23A* and decreased expression of several T cell chemokines (*CCL4*, *CCL5*, and *CXCL13*), which might dampen the infiltration of conventional effector CD8⁺ T cells and NK cells into recurrent tumors (Harlin et al., 2009; Kortlever et al., 2017; Pedersen et al., 2016). Then, we determined the cellular signatures, normalized against the top 50 upregulated genes associated with *KLRB1*^{high} CD8⁺ T cells (STAR methods), to stratify 39 early recurrent patients from validation cohort 2 into high or low groups based on bulk RNA-seq. The high group correlated with significantly poorer clinical prognosis (Figure S5D). Analyses of primary HCC cohorts from TCGA (The Cancer Genome Atlas Research Network, 2017) also supported a *KLRB1*^{high} CD8 T cell signature as a significant indicator of adverse clinical outcomes in HCC patients (Figure S5E).

In conclusion, CD8⁺ T cells in RT samples, characterized by overexpression of *KLRB1*, demonstrated an innate-like dysfunctional state, with low cytotoxicity and immunosuppressive phenotypes, which differed from the exhaustion state observed in PT samples. Notably, the enrichment of the gene expression signature associated with *KLRB1*^{high} CD8 T cells correlated with poor prognosis and tumor progression in both relapse and primary HCC.

CD8⁺ T cells show reduced clonal expansion in early-relapse HCC

Next, we examined the clonal expansion capacity of CD8⁺ T cells. T cells that shared the same TCR clones with more than one T cell in the same patient were defined as clonal T cells, and their increase was indicative of clonal expansion. We observed that RT and PT samples had similar abundances of clonal CD8⁺ T cells (mean: 0.42 versus 0.39; Figure S6A). However, when we analyzed the proliferative capacities of those

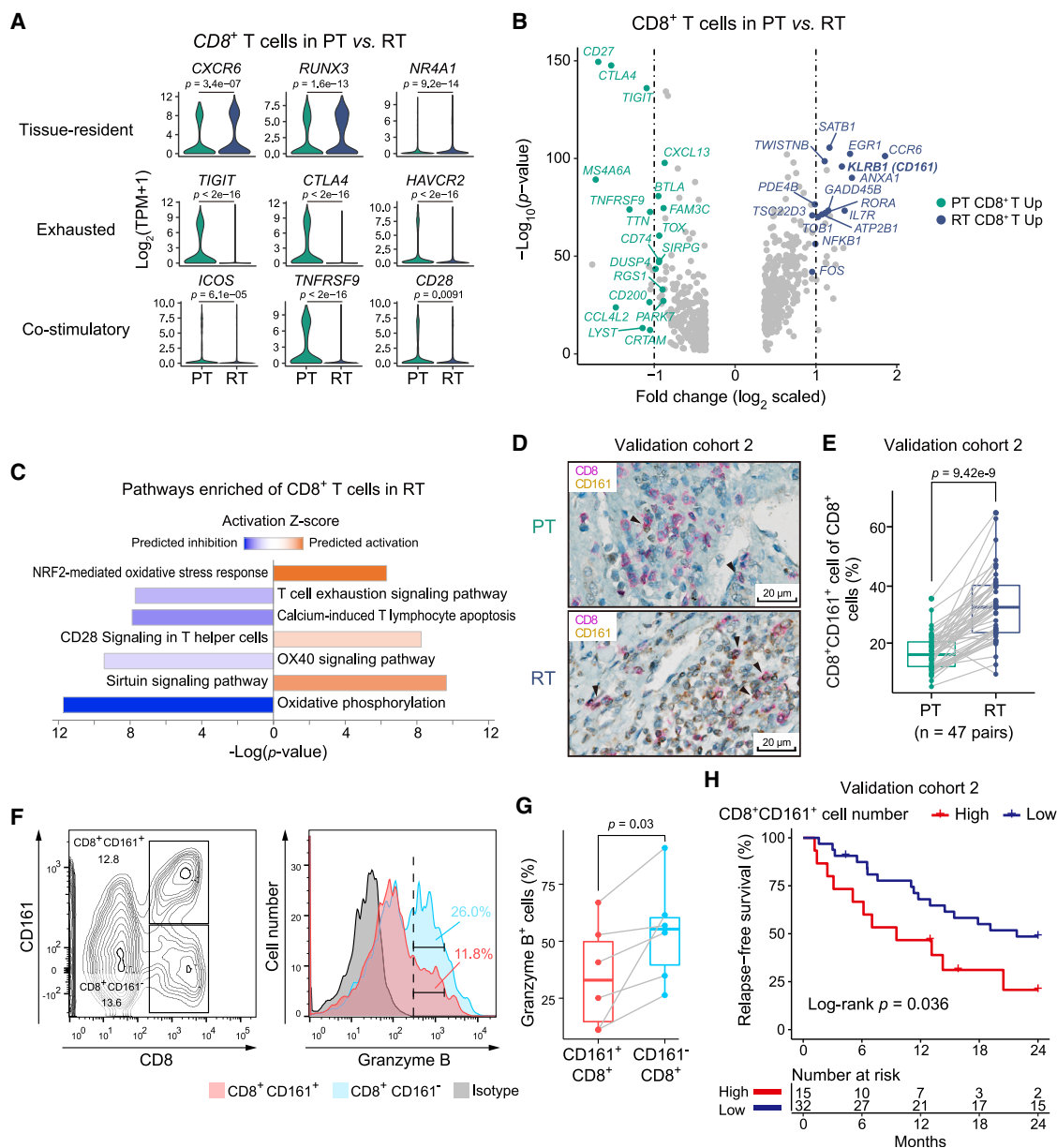


Figure 5. $CD8^+$ T cells in relapsed HCC reside in an innate-like phenotype

(A) Violin plot indicating the expression of tissue-resident, exhausted, and co-stimulatory genes in $CD8^+$ T cells in PT (green) and RT (blue) samples. The p values were calculated by Student's t test.

(B) Volcano plot shows differentially expressed genes between PT (green dots) and RT $CD8^+$ T cells (blue dots). The names of the most significant genes are indicated in the plots.

(C) Two-sided bar graph showing the enriched activated and inhibited pathways in $CD8^+$ T cells in RT, by IPA.

(D) IHC staining of CD8 and CD161 antibodies, showing the infiltration of $CD8^+CD161^+$ T cells, in paired PT and RT samples from the validation cohort 2. Scale bars, 20 μ m.

(E) Boxplot illustrating the fraction of $CD8^+CD161^+$ T cells, in 47 matched PT and RT pairs, based on IHC results. Statistical analyses are by paired Wilcoxon test.

(F) Two-dimensional FACS plot showing $CD8^+CD161^+$ and $CD8^+CD161^-$ T cells from HCC tumor tissue (left panel). The distribution of granzyme B expression in these cells is represented as an overlaid histogram (right panel). The proportion of cells is indicated in the plot.

(G) Boxplot indicating the proportions of GZMB⁺ cells in $CD8^+CD161^-$ and $CD8^+CD161^+$ T cells. The p values are from a paired Wilcoxon test.

(H) Kaplan-Meier analysis showing the relapse-free survival rate of patients, characterized by either low (blue) or high (red) numbers of $CD8^+CD161^+$ T cells. The numbers of patients and the risk classification are indicated in the figure. Significance was calculated using the log-rank test.

N represents the number of patients.

See also Figure S5 and Tables S4 and S5.

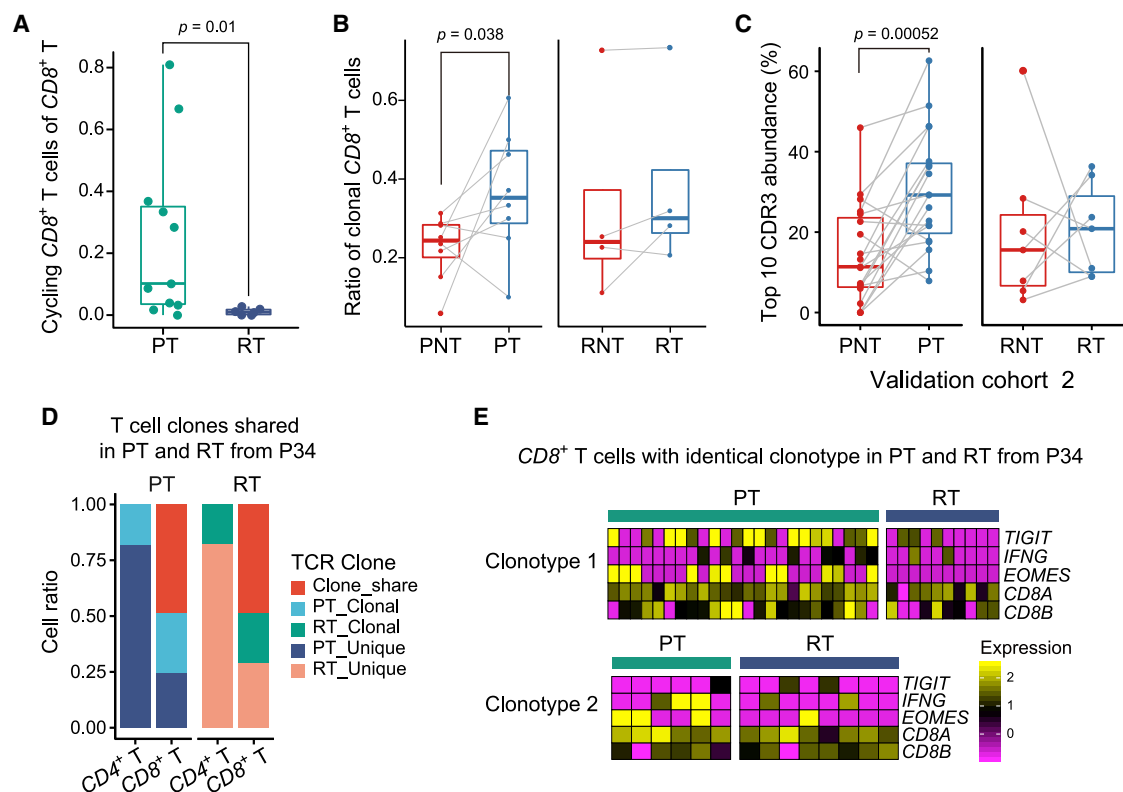


Figure 6. CD8⁺ T cells show reduced clonal expansion in relapsed tumors

(A) Boxplots showing the fraction of cycling CD8⁺ T cells in PT (green) and RT (blue) samples. The p value was calculated by the Wilcoxon test.

(B) Boxplots showing the fractions of CD8⁺ clonal T cells in PNT, PT, RNT, and RT samples, based on single-cell RNA-seq. The p value was calculated by the paired Wilcoxon test.

(C) Boxplots showing the abundance of the top T cell clones represented by complementarity-determining region 3 (CDR3) in PNT, PT, RNT, and RT samples, based on TCR-seq.

(D) Stacked bar plot, showing the TCR clone distribution in CD4⁺ and CD8⁺ T cells. The shared TCR clones between RT and PT samples (Clonal_share), unique clones in RT samples (Clonal_R), unique clones in PT samples (Clonal_P), and unique TCRs (Unique) are labeled using different colors.

(E) Heatmap showing the expression of selected genes in T cells belonging to the same clone from RT and PT of patient P34. Two clones are shown on the upper and lower panel, respectively. The statistical analysis was performed using a paired Wilcoxon test.

See also Figure S6.

cells, we observed that few CD8⁺ T cells in RT samples were in a cycling state (Figure 6A). A higher fraction of CD8⁺ T cell clones were shared between RT and RNT samples than those shared between PT and PNT samples (Figure S6B), revealing that adjacent non-tumor tissues serve as reservoirs that can provide more CD8⁺ T cells for infiltration into tumor sites in RT. However, the proportion of clonal CD8⁺ T cells did not change between RT and RNT samples, whereas a significant increase in PT was observed compared with PNT (Figure 6B). We further confirmed that the top ten abundant T cell clones represented by complementarity-determining region 3 (CDR3) in tumors were increased in PT compared with PNT samples, but no difference was observed between RT and RNT samples, based on bulk T cell receptor sequencing (TCR-seq) (Figure 6C; STAR methods). In conclusion, CD8⁺ T cells did not show clonal expansion in RT samples.

In the paired samples from patient P34, 67% of clonal CD8⁺ T cells in the RT shared the same TCRs with CD8⁺ T cells in the PT sample (Figure 6D). Among these, most CD8⁺ T cells in

the RT recognized the same antigens as CD8⁺ T cells in the PT sample. The fractions of cells from shared T cell clones between PT and RT of total CD8 T cells were similar between RT and PT samples (48% versus 48%), revealing that these T cell clones did not show any apparent expansion from PT to RT. Additional gene expression profile comparisons among CD8⁺ T cells belonging to the same clones in RT and PT samples demonstrated that the dysfunctional, non-activated CD8⁺ T cell state in RT samples derive from clones that arising in the PT samples, indicating that those cells may represent central memory cells (Amsen et al., 2018) (Figure 6E). Moreover, we observed that malignant cells in RT originated from a minor clone of PT cells by single-cell DNA-seq (Figure S6C; STAR methods), which revealed the loss of the major clone of malignant cells during recurrence. Besides, we also found that at least one major clonal mutation (cancer cell fraction [CCF] >0.8) from the primary tumor was nearly lost (CCF <0.05) in the paired clonal related relapsed tumor in 22 of 24 HCC patients from an independent cohort (Ding et al., 2019), indicating the loss of major neoantigens is a

common phenomenon in early-relapse HCC. Taken together, this non-activated phenotype of CD8⁺ T cells in RT samples might be due to the loss of matching neoantigens (Reading et al., 2018) in recurrent tumor cells.

Our data suggest a model in which CD8⁺ T cell clones reside in a low proliferative and unresponsive state in RT due in part to tumor selection, suggesting that those cells are unable to recognize and eliminate recurrent tumor cells displaying sub-clonal neoantigens.

The contribution of malignant cells to the unique early-relapse HCC ecosystem

Our analyses demonstrated that relapsed HCC was associated with reduced proportions of Tregs and no increase in macrophages or other immune-suppressive T cell subtypes. This led us to hypothesize that the substantial differences in the tumor ecosystem between PT and RT could be caused by differences in malignant cells. To resolve these differences, we performed a DEG analysis between PT and RT malignant cells from the discovery cohort. We observed an enrichment of genes involved in cell-cycle-related pathways (e.g., *MYC_targets_V1* and *G2M_checkpoint* pathways) in PT, whereas the genes upregulated in RT primarily belonged to immune response pathways (e.g., *INF_gamma* signaling and *TNFA_signaling_via_NFKB*) (Figure 7A; Table S6). We also observed that PT samples had higher proliferative and lower hepatic function signals, whereas RT samples presented higher immune surveillance and escape signatures, which were confirmed in validation cohort 1 (Figures 7B and S7A; Table S3; STAR methods). Among the genes related to immune escape, we observed increased *CD47* expression in malignant cells from RT, suggesting the potential inhibition of DC maturation and antigen-presentation (Xu et al., 2017), as well as resistance against macrophage targeting (Okazawa et al., 2005; Tsai and Discher, 2008), with a concomitant reduction in neoantigen release. Similarly, we observed increased expression levels of *HLA/B/C* in RT, indicating an increased resistance to NK-mediated cell death (Figure 7C). Patient P07 carried an *HLA-B* missense mutation in the RT sample that was not found in the corresponding PT sample, potentially resulting in decreased neoantigen peptide presentation in malignant cells. Our data demonstrate that RT possesses low proliferation rates and a high capacity to evade immune-mediated cell death.

We also examined the expression of immune checkpoint ligands and receptors, observing that many genes, such as *LGALS9* (ligand for TIM-3), *TNFRSF14* (ligand for BTLA), *PDCD1LG2* (ligand for PD-1), and *VTCN1* (ligand for B7-H1) displayed low expression levels in both primary and relapsed malignant cells (Figure S7B). In contrast, *FGL1*, which encodes a ligand for LAG3, was abundant in both PT and RT (Figure S7B). Moreover, malignant cells from RT showed significantly upregulated *CD274* (PD-L1) expression and a slightly higher expression of *CTLA4*, compared with PT (Figure 7D). Similarly, a higher proportion of *CD274*⁺ malignant cells was observed in RT, whereas RT and PT had similar fractions of *CTLA4*⁺ malignant cells (Figures 7E and S7C). However, CD8⁺ T cells in RT harbored reduced exhaustion states compared with those from PT (Figures S5A–S5C). We postulated that recurrent malignant cells that overexpressed *CD274* or *CTLA4* might induce compro-

mised anti-tumor immunity via an alternative method other than classic inhibitory signals. Ligand-receptor (L-R) analysis suggested crosstalk between malignant cells and DC2/DC3 cells, via *CD274/CD80* and *CTLA4/CD80* interactions in RT but not PT (Figure 7F). Due to the higher affinity of CD80 for PD-L1 and CTLA4 than for CD28 (Butte et al., 2007), PD-L1⁺ or CTLA4⁺ malignant cells may reduce the efficiency of antigen-presentation and the activation of T cells (Chen et al., 2017; Kowalczyk et al., 2014), which usually occurs through CD80-CD28 interactions between DCs and T cells, during tumor recurrence. We also performed multiplexed immunofluorescent staining to provide evidence that CD80⁺CD1C⁺ DCs reside near PD-L1⁺ relapsed malignant cells, supporting their crosstalk (Figure 7G). Moreover, a recent study demonstrated that tumor cells released exosomal PD-L1 to compromise antitumor immunity (Chen et al., 2018a).

To further investigate the interactions that occur in the ecosystem, we utilized significant L-R pairs to calculate the strengths of the interactions. Malignant cells showed close interactions with endothelial, CD4 CCR7, and pDC clusters in PT; however, they showed strong regulatory relationships with CD8 NR4A1, CD8 CCR6, and DC1 subtypes in RT (Figure S7D). We found that recurrent malignant cells showed the strongest chemokine interactions with CD8 NR4A1 and CD8 CCR6 clusters, via the CCL20-CCR6 axis, suggesting that malignant cells attracted innate-like CD161⁺CD8⁺ T cells (Jeffery et al., 2017), via CCL20 in RT (Figure S7E).

In summary, our data demonstrated that malignant cells in the RT exhibited enhanced immune evasion capacities and reduced cellular proliferation. Recurrent malignant cells could compromise antigen presentation in DC cells, via the PD-L1-CD80 and CTLA4-CD80 axes, and to recruit innate-like CD161⁺ CD8⁺ T cells via the CCL20-CCR6 axis, which may shape dysfunctional anti-tumor immunity in early-relapse HCC.

DISCUSSION

Despite the progresses in surveillance and treatment strategies, which have improved overall survival rates, the clinical outcomes of HCC remain dismal due to the high incidence of early recurrence, even after surgical resection (Bosch et al., 2004; Rahbari et al., 2011). Recurrent tumors are often treated based on the molecular and pathological features of the primary tumor, although whether they share similar microenvironment characteristics has not been evaluated. A recent study indicated that primary and early-relapse HCC have similar genomic alterations (Ding et al., 2019). However, differences in the cellular ecosystems between primary and recurrent HCC remain to be deciphered. Here, we provide a comprehensive single-cell transcriptomic atlas to characterize tumor ecosystems in primary and early-relapse HCC. Our analysis reveals a distinct immune ecosystem in early-relapse HCC, characterized by reduced fractions of classic immunosuppressive cells Tregs and the increased fractions of DCs and innate-like CD161⁺CD8⁺ T cells, with dysfunctional cytotoxicity and low expansion (Figure 7H). Transcriptomic profiling, augmented by L-R based cell-cell interaction analysis, illustrates recurrent malignant cell crosstalk with DCs and T cells, which may compromise

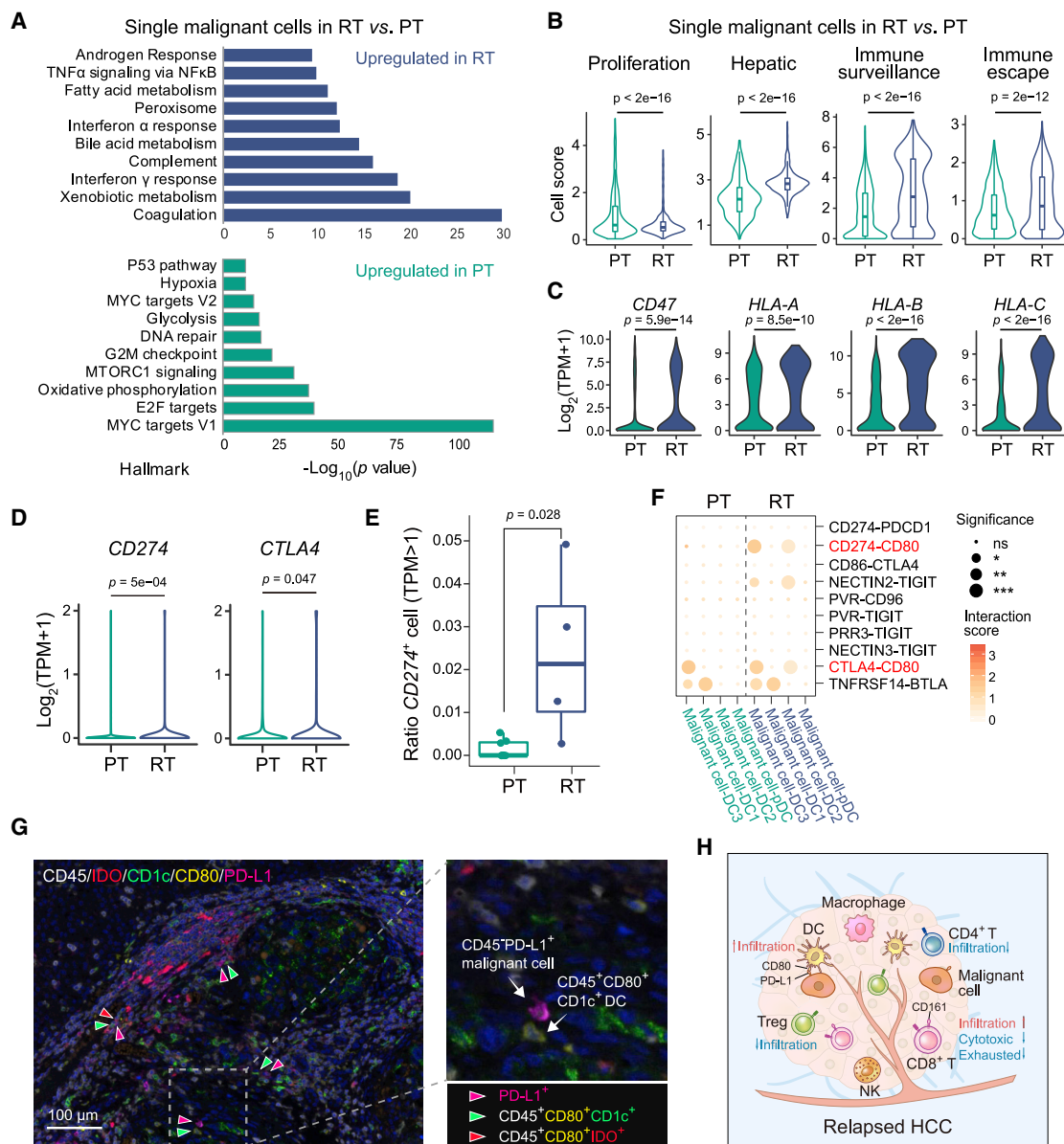


Figure 7. Characteristics of malignant cells and their effects on the tumor ecosystem

(A) Bar chart showing the enrichment of specific pathways, based on the HALLMARK gene set of upregulated genes, in primary or recurrent malignant cells. (B) Violin plots showing the proliferation, hepatic scores, immune escape, and immune surveillance scores of malignant cells from PT (green) and RT (blue) samples. (C) Violin plot showing the expression of selected immune escape genes in tumor cells from PT (green) and RT (blue) samples. (D) Violin plot showing the expression of *CTLA4* and *CD274* in primary and relapsed malignant cells. (E) Boxplot showing the fraction of *CD274*⁺ tumor cells in RT and PT samples. (F) Bubble chart showing the interaction between malignant cells and DC subtypes, based on selected ligand and receptor pairs. These scores are normalized expression levels, and the sizes of the bubbles indicate the significance of the interactions, calculated by CellPhoneDB. (G) Immunofluorescence images showing the interaction between DCs and malignant cells, based on the CD80-PDL1, using antibodies CD45, IDO, CD1c, cd80, and PD-L1. Scale bars, 100 μ m. (H) Schematic diagram of the unique tumor ecosystem in relapsed HCC. The p values in (B–D) were calculated by Student's t test, and in (E) by the Wilcoxon test. See also Figure S7 and Tables S3 and S6.

antitumor immunity (Figure 7H). Our study represents an essential step toward understanding how the immune environment shapes tumor recurrence and reveals the existence of active

crosstalk between recurrent malignant cells and immune cells in HCC, facilitating the escape of tumor cells from anti-tumor immunity.

In primary HCC, the landscape of infiltrating T cells is characterized by Tregs enrichment, leading to an immunosuppressive microenvironment (Zheng et al., 2017a). Here, we reveal a distinct T cell composition in early-recurrent HCC, characterized by the exclusion of Tregs and enrichment of CD8⁺ T cells. Compared with CD8⁺ T cells from PT, CD8⁺ T cells in RT possessed remarkably different transcriptional characteristics, with significantly decreased expression of exhaustion, proliferation, and cytotoxicity signatures and elevated expression of tissue-residence genes. Moreover, DEG analysis suggested that CD8⁺ T cells derived from RT were characterized by the overexpression of *KLRB1* (CD161), *KLRG1*, *NCR3*, *RORA*, and *IL7R*, suggesting an innate-like and memory phenotype (Chiossone et al., 2018; Fergusson et al., 2014; Joshi et al., 2007). In microbial infection settings, *KLRB1*^{high} CD8 T cells comprise a pool of resting memory cells, expressing both *KLRG1* and *IL7R*, which contribute to the early control of microbial recall infections caused by the same pathogen, in both antigen-dependent and -independent manners (Obar et al., 2011; Ruiz et al., 2014). The adaptive immune response of memory CD8⁺ T cells is considered a central component of the immune response to the disseminated disease (Amsen et al., 2018). However, early recurrence developed, regardless of the significant elevation of CD161⁺CD8⁺ T cell infiltration, suggesting that these T cells are unable to prevent the intrahepatic dissemination of HCC. We also demonstrate a significantly lower production of granzyme B by CD161⁺CD8⁺ T cells than CD161[−]CD8⁺ T cells, supporting their poor anti-tumor function. TCR analysis indicated that CD8⁺ T cells in RT show limited clonal expansion and share most of TCR clone types with their counterparts in PT. This finding implies that CD161⁺CD8⁺ T cells in RT could mount an efficient immune response, targeting clonal neoantigens in primary tumors but not effectively activated in RT. Further analysis of sequencing data from paired primary and recurrent tumor cells revealed that the origin of early-relapse tumor cells was subclonal. Therefore, minor subclones, rather than major clones, fuel early recurrence, and become the dominant clones in recurrent tumors (Angelova et al., 2018). The loss of major clonal mutations of the primary tumor in the corresponding clonal-related relapsed tumor is in agreement with recent data (Ding et al., 2019). This loss of major clonal neoantigens during recurrence may explain why CD161⁺CD8⁺ T cells with memory phenotypes cannot prevent recurrence, because they are unable to recognize minor subclonal neoantigens in RT. Moreover, *KLRB1*^{high} CD8 T cells downregulated T cell recruitment chemokines (*CCL4*, *CCL5*, and *CXCL13*) and upregulated the exclusion chemokine (*IL23A*), dampening the infiltration of functional effector CD8⁺ T cells into recurrent tumors (Harlin et al., 2009; Kortlever et al., 2017; Pedersen et al., 2016).

DCs excel at antigen presentation and play a vital role in priming anti-tumor T cell immunity. In our study, the increased infiltration of DCs was not associated with boosted effector functions of CD8⁺ T cells in RT, implicating a compromised antigen presentation function in RT. Co-signaling pathways in the B7-CD28 family provide critical signals that regulate T cell activation and inhibition (Chen, 2004). We observed that relapsed HCC exhibited a higher proportion of CD274⁺ (PD-L1) malignant cells than primary tumors. Due to the increased affinity

of CD80 for PD-L1 compared with CD28 (Butte et al., 2007), CD80 on a DC might preferentially bind to PD-L1 on recurrent tumor cells rather than CD28 on a resting T cell. The multicolor immunofluorescence staining of early-relapse HCC confirmed the physical juxtaposition of PD-L1-expressing tumor cells and CD80-expressing DCs. Therefore, one potential immune evasion mechanism of early-recurrent HCC could involve PD-L1⁺ tumor cells, reducing CD80-CD28 interactions through the competitive binding of PD-L1 to CD80, which, in turn, prevents T cell activation from killing tumor cells. Concerning the inability of CD161⁺CD8⁺ T cells to perform a recurrent tumor-specific cytolytic attack, induction of neoantigen release from relapsed tumors by chemotherapy or targeted therapy might activate naive CD8⁺ T cells and lead to clonal expansion of recurrent tumor-specific CD8⁺ T cells (Peng et al., 2019). In contrast, PD-L1 blockade could replenish the co-stimulatory signaling pathways between DC and CD8 T cells, which is compromised by PD-L1 expression on recurrent tumor cells. Together with emerging clinical trial data, our results suggested that tumor immunotherapy, combined with targeted therapies or chemotherapy, might be a more rational treatment approach than an ICB as a single agent for patients with recurrent HCC.

The different efficacies of ICB therapies among patients and cancer types suggest the need to identify which patients are likely to benefit from ICBs and establish optimal immunotherapeutic strategies, based on a better understanding of the heterogeneity of the immune checkpoint expression in the primary and recurrent/metastatic HCC microenvironment. ICB therapy resulted in only 16%–20% objective response rates among advanced HCC patients (Cheng et al., 2020). Our data revealed that *PDCD1* (PD-1) was expressed in a small proportion of T cells, and *HAVCR2* and *CTLA4* displayed higher expression, indicating that TIM-3 or CTLA-4 blockades might be of more significant benefit for HCC treatment. These findings may explain, at least in part, the better response rates observed in clinical trials that use a combination of PD-1 and CTLA-4 inhibitors (CheckMate-040 trial; 31% patients responding to treatment) (Yau et al., 2019), compared with those using PD-1 inhibitors alone (KEYNOTE-224 trial; 17.3% of patients responding to treatment) (Zhu et al., 2018). More importantly, we found the reduced expression of immune checkpoint genes *CTLA4*, *HAVCR2*, and *TIGIT* in T cells from relapsed tumors, suggesting that checkpoint blockade approaches that target these markers might be effective for primary HCC, but not in recurrent patients. Recurrent tumor cells presented the increased expression of CD47, an antiphagocytic “don’t eat me” signal. A CD47 blockade inhibits tumor growth in mouse models of HCC (Lo et al., 2016; Xiao et al., 2015). These results suggest that different therapeutic approaches are required to treat primary or relapsed HCC, and anti-CD47-based therapies might represent a feasible prophylactic option for the early recurrence of HCC, whereas the combination of multiple ICBs could be beneficial for a larger HCC population.

In conclusion, this study provides evidence of the tumor ecosystem heterogeneity between primary and early-relapse HCC, in terms of immune and tumor phenotypes, T cell developmental trajectories, TCR clonality, and unique signatures, in

addition to the crosstalk between tumor and immune cells. Our data can be a valuable resource, facilitating a deeper understanding of the mechanisms associated with early recurrence and assisting in developing more effective therapeutic targets and biomarkers for immunotherapies in HCC patients.

STAR★METHODS

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SUPPLEMENTAL INFORMATION

Supplemental Information can be found online at <https://doi.org/10.1016/j.cell.2020.11.041>.

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AUTHOR CONTRIBUTIONS

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DECLARATION OF INTERESTS

The authors declare no competing interests.

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies		
Anti-human CD45 antibody (Clone HI30)	BD Biosciences	Cat# 555485; RRID: AB_398600
Anti-CD8 alpha antibody (Clone C8/144B)	Abcam	Cat# ab17147; RRID: AB_443686
Anti-CD161 antibody	Abcam	Cat# ab197979
CD45 (D9M8I) XP Rabbit mAb	Cell Signaling Technology	Cat# 13917; RRID: AB_2750898
PE anti-human CD8a antibody (Clone HIT8a)	BioLegend	Cat# 300908; RRID: AB_314112
APC/Cyanine7 anti-human CD161 antibody (Clone HP-3G10)	BioLegend	Cat# 339928; RRID: AB_2563967
FITC anti-human/mouse Granzyme B recombinant antibody	BioLegend	Cat# 372206; RRID: AB_2687030
IDO (D5J4E) Rabbit mAb	Cell Signaling Technology	Cat# 86630; RRID: AB_2636818
CD45 Recombinant Rabbit Monoclonal Antibody (Clone RR691)	Biolynx	Cat# BX00087
PD-L1 (E1L3N) XP Rabbit mAb	Cell Signaling Technology	Cat# 13684; RRID: AB_2687655
Human B7-1/CD80 Antibody (Clone # 37711)	R&D Systems	Cat# MAB140; RRID: MAB140
CD1C Mouse Monoclonal Antibody (Clone OT1H9)	Origene	Cat# TA505384; RRID: AB_2623048
Biological samples		
HCC samples	Zhongshan Hospital Fudan University	N/A
Adjacent normal tissues from HCC patients	Zhongshan Hospital Fudan University	N/A
Chemicals, peptides, and recombinant proteins		
Fluorescein Diacetate (FDA)	Thermo Fisher Scientific	Cat# F1303
Human CD3/CD28 T Cell Activator	STEMCELL	Cat# 10971; RRID: AB_2827806
BD Golgi plug Protein Transport Inhibitor	BD Biosciences	Cat# 555029; RRID: AB_2869014
4',6-diamidino-2-phenylindole (DAPI)	Sigma-Aldrich	Cat# D9542
Critical commercial assays		
Human Tumor Dissociation Kit	MiltenyiBiotec	Cat# 130-095-929
Human Tumor Cell Isolation Kit	MiltenyiBiotec	Cat# 130-108-339
CD8 Microbeads	MiltenyiBiotec	Cat# 130-045-201
Cytofix/Cytoperm Fixation/Permeabilization Kit	BD Biosciences	Cat# 554714
Qubit dsDNA BR Assay Kit	Thermo Fisher Scientific	Cat# Q32850
Qubit RNA HS Assay	Thermo Fisher Scientific	Cat# Q32855
AllPrep DNA/RNA Mini Kit	QIAGEN	Cat# 80204
Allprep DNA/RNA FFPE extraction kit	QIAGEN	Cat# 80234
MGEasy Exome Universal Library Prep Set	MGI	Cat# 1000009657
MGEasy Circularization Kit	MGI	Cat# 1000005259
Single Cell 5' Library Construction Kit	10x Genomics	Cat# 1000020
Single Cell V(D)J Enrichment Kit, Human T Cell	10x Genomics	Cat# 1000005
TruSeq RNA Access Library Prep Kit	Illumina	Cat# RS-301-2001

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REAGENT or RESOURCE	SOURCE	IDENTIFIER
Deposited data		
Processed expression data files for human scRNA-seq	This paper	CNSA: CNP0000650
Processed scRNA-seq data of PBMC-derived DC	Villani et al., 2017	GEO: GSE94820
Processed scRNA-seq data of head and neck cancer	Puram et al., 2017	GEO: GSE103322
Software and algorithms		
SOAPnuke (v1.5.6)	Chen et al., 2018c	RRID: SCR_015025; https://github.com/BGI-flexlab/ SOAPnuke
Bowtie2 (v2.3.2)	Langmead and Salzberg, 2012	RRID: SCR_005476; http://bowtie-bio.sourceforge.net/bowtie2/index.shtml
RSEM (v1.3.0)	Li and Dewey, 2011	RRID: SCR_013027; https://github.com/deweylab/RSEM
Picard (v1.73)	Broad Institute	RRID: SCR_006525; https://github.com/broadinstitute/picard
Genome Analysis Toolkits (v3.8.1)	McKenna et al., 2010	RRID: SCR_001876; https://gatk.broadinstitute.org
ANNOVAR (version Feb 2016)	Wang et al., 2010	RRID: SCR_012821; http://annovar.openbioinformatics.org/en/latest/
inferCNV (v1.2.1)	Patel et al., 2014	https://github.com/broadinstitute/inferCNV
ABSOLUTE (v1.0.6)	Carter et al., 2012	RRID: SCR_005198; https://www.broadinstitute.org/cancer/cga/ABSOLUTE
PhyloicNDT	Leshchiner et al., 2018	https://github.com/broadinstitute/PhyloicNDT
Seurat (v2.4.3)	Satija et al., 2015	RRID: SCR_016341; https://github.com/satijalab/seurat
TraCeR (v0.3.0)	Stubbington et al., 2016	RRID: SCR_016338; https://github.com/Teichlab/tracer
Monocle2 (v2.10.1)	Qiu et al., 2017	RRID: SCR_016333; https://github.com/cole-trapnell-lab/monocle-release
GSVA (v1.36.0)	Hänzelmann et al., 2013	http://www.bioconductor.org/packages/release/bioc/html/GSVA.html
GSEA (v6.0)	Subramanian et al., 2005; Mootha et al., 2003	RRID: SCR_003199; https://www.gsea-msigdb.org/gsea/index.jsp
Ingenuity Pathway Analysis (IPA)	Ingenuity	RRID: SCR_008653; https://digitalinsights.qiagen.com/products-overview/discovery-insights-portfolio/content-exploration-and-databases/qiagen-ipa/
CellPhoneDB (v2.0.3)	Vento-Tormo et al., 2018	RRID: SCR_017054; https://github.com/Teichlab/cellphonedb
Cell Ranger (v3.0.1)	10x Genomics	RRID: SCR_017344; https://10xgenomics.com
iMonitor (v1.3.0)	Zhang et al., 2015	https://github.com/zhangwei2015/IMonitor

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REAGENT or RESOURCE	SOURCE	IDENTIFIER
Other		
Interactive explorer of cells from HCC patients	This paper	https://db.cngb.org/PRHCCdb/

RESOURCE AVAILABILITY**Lead contact**

Further information and requests for resources and reagents should be directed to and will be fulfilled by the Lead Contact, Jia Fan (fan.jia@zs-hospital.sh.cn).

Materials availability statement

This study did not generate new unique reagents.

Data and code availability

The accession number for the processed expression data of RNA-seq reported in this paper is CNP0000650 (<https://db.cngb.org/search/project/CNP0000650>). The data can also be explored at <https://db.cngb.org/PRHCCdb>. The original data of PBMC DCs and head and neck cancer for Figures S2E–S2H in this paper are available with accession number GSE94820 and GSE103322 (Puram et al., 2017; Villani et al., 2017). All other data are available upon reasonable request for Lead Contact/corresponding author.

EXPERIMENTAL MODEL AND SUBJECT DETAILS**Human subject**

A total of 69 patients were pathologically diagnosed with HCC at Zhongshan Hospital Fudan University and were enrolled in this study. Paired, fresh tumor (2 × 2 × 1 cm) and adjacent normal tissue samples (at least 2 cm from matched tumor tissues) were collected as the Discovery cohort, from 18 treated naive primary or early-relapse HCC patients, detailed clinical and pathological information including age, gender, tumor size, prognosis et al. shown in Table S1. Frozen tumor tissues or fresh cell suspensions from pairwise primary and relapsed tumor samples were collected from 4 patients as Validation cohort 1. Formalin-fixed paraffin-embedded (FFPE) tissue blocks of pairwise primary and early relapsed tumors were collected from 47 patients as Validation cohort 2. The available clinical features of Validation cohorts are summarized in Table S7. All patients gave informed consent for collection of clinical information, tissue collection, research testing under Institutional Review Board (IRB)-approved protocols (B2017-159R and B2019-060R) at Zhongshan Hospital Fudan University.

METHOD DETAILS**Single-cell isolation**

Fresh tumor and adjacent normal tissue samples were surgically removed from patients and immersed in a complete medium containing 90% Dulbecco's modified eagle medium (DMEM; Cat# 11054001, GIBCO) and 10% fetal bovine serum (FBS; Cat# 16140071, GIBCO), and transported to the lab in a refrigerated container. Suitable small tissue blocks (no necrotic foci / no hemorrhagic foci / less fibrous connective tissue) were cut into pieces (diameter 1–3 mm). The pieces were transferred to the gentle MACS C Tubes (Cat# 130-096-334, Miltenyi Biotec), with 5 mL of digestive enzyme included in Tumor Dissociation Kit (Cat# 130-095-929, MiltenyiBiotec). Then the tissues were made into single-cell suspension using the gentleMACS Dissociator (Cat# 130-093-235, MiltenyiBiotec) following these steps: milled; incubated at 37°C for 30 min on a shaker; milled; incubated at 37°C for 30 min; milled; filtered through a 70 μm filter, plus 2% FBS. Finally, the single-cell suspension was centrifuged at 400 g for 7 mins, resuspended in CELLSAVING (Cat# C40050, NCM biotech), cryopreserved in Cryotubes (Cat# 430488, Corning) and then stored at –80°C.

For the paired Validation Cohort 1, single-cell or single-nucleus (sn) were used for scRNA-seq or snRNA-seq. The preparation of single-cell suspension was the same as described in last paragraph. Single-nucleus were prepared as following: frozen tumor tissue pieces were placed in 1 mL Homogenization Buffer (pre-chilled) containing 10mM Tris pH 8.0 (Cat# AM9856, Ambion), 250 mM sucrose (Cat# V900116-500G, Sigma-Aldrich), 25 mM KCl (Cat# AM9640G, Ambion), 5 mM MgCl₂ (Cat# AM9530G, Ambion), 1% bovine serum albumin (BSA; Cat# C102301, Sangon Biotech), 0.1% NP-40 (Cat# 11754599001, Roche), 0.2U/μl RNasin Plus (Cat# N2611, Promega), 1 × protease inhibitor (Cat# G6521, Promega), and 0.1mM DTT (Cat# 707265ML, Thermo Fisher Scientific) in 2 mL dounce homogenizer (Cat# D8938, Sigma-Aldrich) for 5 minutes. The tissue was then homogenized by 10 strokes of the loose pestle and 10 strokes of the tight pestle. The Dounce homogenizer was submerged in ice during the grinding step. 1ml Homogenization Buffer was added to the Dounce homogenizer then the homogenate was passed through 40 μm cell strainer (Cat# 43-10040-70, Miltenyi Biotec) into a 15 mL conical tube and centrifuged at 300 g × for 5 mins. Sediment was resuspended in 1.5 mL Blocking

Buffer containing 1 × phosphate buffer saline (PBS; Cat# 10010049, Thermo Fisher Scientific), 1% filtered sterilized BSA, and 0.2U/μl Rnasin Plus by pipetting up and down gently on ice and centrifuged at 300 g × for 5 mins. The previous step was repeated once. Nuclei were resuspended in 0.04% BSA of PBS then stained by 4′-6′-diamidino-2-phenylindole (DAPI; Cat# D9542, Sigma-Aldrich) for counting and diluted in an appropriate concentration.

Single cell sorting, reverse transcription, amplification and sequencing

The cryopreserved single-cell suspension was incubated in a 37°C water bath for rapid recovery and washed twice with 1 × PBS. Then the single-cell suspension was stained with antibodies against CD45 (Cat# 555485, BD) and fluorescein diacetate (FDA; Cat# F1303, Thermo Fisher Scientific) for FACS sorting. Single living immune cells (CD45⁺, FDA⁺) and non-immune cells (CD45[−], FDA⁺) were obtained after FACS (Figure S1B). Then full-length single-cell RNA preparation in microwell was performed using MIRALCS (Wu et al., 2015), a modified microwell-based SMART-seq2 (Picelli et al., 2014) method. Briefly, using the following steps to prepare cDNA products: dispensing lysis buffer and single-cell into 5,184 microwell chip, cell lysis, reverse transcription, cDNA amplification and identify single-cell wells based on qPCR results, product extraction, fragmentation and PCR, and then cyclized into ssDNA libraries. Each single-cell library was labeled with a barcode in the PCR step, and then was sequenced on a BGISEQ500 sequencer with 100-bp single-end reads.

For Validation cohort 1, single-cell or single-nucleus transcriptome amplifications, cDNA libraries and TCR VDJ libraries were prepared using 10x Genomics Chromium Single Cell 5′ Library Construction Kit (Cat# 1000020, 10x Genomics) and Chromium Single Cell V(D)J Enrichment Kit, Human T Cell (Cat# 1000005, 10x Genomics) following the manufacturer's instruction. Briefly, these steps include cDNA amplification reaction, enrichment of the TCR VDJ fragment twice from the cDNA product and the library construction of cDNA or TCR VDJ. Finally, the libraries were cyclized into ssDNA libraries by MGIEasy Circularization Kit (Cat# 1000005259, MGI). The cDNA libraries were sequenced on the BGISEQ500 sequencer with 8+26+100-bp pair-end reads and the TCR VDJ libraries were sequenced on the BGISEQ500 sequencer with 8+150+150-bp pair-end reads.

Bulk DNA and RNA extraction and sequencing

Genomic DNA and RNA of tissue samples were extracted using the QIAamp AllPrep DNA/RNA Mini Kit (Cat# 80204, QIAGEN) according to the manufacturer's specifications including the steps of DNA/RNA adsorption, purification, and collection in columns. The concentrations of DNA were quantified using Qubit dsDNA BR Assay Kit (Cat# Q32850, Thermo Fisher Scientific) and the quality of the DNA was evaluated by agarose gel electrophoresis. The exome libraries were constructed using the MGIEasy Exome Universal Library Prep Set (Cat# 1000009657, MGI) according to the instructions. In brief, DNA is fragmented, followed by the steps of adaptor ligation at both ends and PCR amplification. Samples were sequenced on a BGISEQ500 sequencer with 100-bp paired-end reads. In addition, multiplex PCR was designed to amplify CDR3 regions of the rearranged TCR β chain (TRB) from genomic DNA. The assay utilized a suite of 32 TRB-V primers that were each specific for a functional VB segment and 13 TRB-J primers for the JB segment (Zhang et al., 2015). The enriched TRB products were cyclized into ssDNA libraries by MGIEasy Circularization Kit, and then were sequenced on the BGISEQ500 sequencer with 100+10-bp single-end reads.

Bulk RNA sequencing of human FFPE samples

For Validation Cohort 2, genomic DNA and total RNA were simultaneously extracted from FFPE slides by means of the AllPrep DNA/RNA FFPE extraction kit (Cat# 80234, QIAGEN), according to the manufacturer's instructions, including the steps of DNA/RNA adsorption, purification, and collection in columns. After purification, DNA and RNA were eluted in 30 μL of nuclease-free water, and yield was determined by the Qubit RNA HS Assay (Cat# Q32855, Thermo Fisher Scientific). A predefined yield of 100 ng of RNA was used as acceptance criteria to ensure adequate library preparation. RNA-seq libraries were constructed using the TruSeq RNA Access Library Prep Kit (Cat# RS-301-2001, Illumina) per manufacturer's recommendations. After library quality and quantity assessment, barcoded libraries were pooled and sequenced by an Illumina HiSeq X Ten instrument (pair-end 150 cycle reactions).

Immunohistochemical staining and evaluation

FFPE tissue blocks from 47 patients (Validation cohort 2) who had undergone resection for recurrent HCC were used for IHC staining. Paired primary and relapsed samples were used for double staining of CD1C (Cat# TA505384, Origene) and IDO (Cat# 86630, CST) antibodies, staining of CD45 (Cat# 13917, CST) antibody, and double staining of CD8 (Cat# ab17147, Abcam) and CD161 (Cat# ab197979, Abcam) antibodies. All the staining process was carried out on the IHC/ISH System (BenchMark GX, Roche) following the manufacturer's instruction.

Single stained cells were counted using a computerized image analysis system (Image-pro plus 6.0, Media Cybernetics Inc.) on the basis of pictures taken under the microscopic fields of 200 × magnification. Double staining samples were evaluated manually under microscopic fields of 1000 × magnification (oil immersion lens). To ensure representativeness and homogeneity, the whole dots were observed in tissue microarray, and 5 fields of immune cell highly infiltrated tumor area were selected in the FFPE block slide. All the evaluation was performed by the same pathologist blinded to patient characteristics.

Tumor infiltrating CD8⁺ T cell functional analysis

Single-cell suspension were obtained from the fresh resection tumor tissue of HCC patient using the protocol described in single-cell isolation section above. Half of the fresh single-cell suspension was processed by Tumor Cell Isolation Kit (Cat# 130-108-339,

Miltenyi Biotec) to separate untouched tumor cells, following manufacturer's protocol. The rest of single-cell suspension was used for sorting CD8⁺ T cell by CD8 Microbeads (Cat# 130-045-201, Miltenyi Biotec). RPMI 1640 (Cat# 22400089, GIBCO) was used for cell culture. For each patient, both homogenized HCC cells and infiltrating CD8⁺ T cells were seeded at the concentration of 2×10^5 cells/mL in the round bottom 96-well plates in triplicate. Then 2 μ L/mL of CD3/CD28 T Cell Activator (Cat# 10971, STEMCELL) was added immediately after seeding. After coculturing for 18 h at 37°C, cells were stimulated with Golgi plug (Cat# 555029, BD Biosciences) protein transport inhibitors for another 6 h in the cell incubator. The 96-well plates were centrifuged at 250 g for 5 mins and the supernatant was discarded. Cells were then resuspended with staining media containing PE-CD8a (Cat# 300908, BioLegend) and APC/Cy7-CD161 (Cat# 339928, BioLegend) antibodies and incubated for 30 mins at room temperature. Cells were washed by staining buffer and pelleted by centrifugation. After using Cytofix/Cytoperm Fixation/Permeabilization Kit (Cat# 554714, BD Biosciences), the thoroughly fixed/permeabilized cells were stained by FITC-Granzyme B (Cat# 372206, BioLegend) antibody in perm/wash buffer at 4°C for 30 mins. Cells were washed and responded in the staining buffer before flow cytometric analysis. Multiparameter FACS data were acquired on the BD FACSAria II instrument (BD Biosciences). After gating of lymphocytes according to FSC and SSC, CD161⁺ CD8⁺ T cell (APC/Cy7⁺ PE⁺) and CD161⁻ CD8⁺ T cells (APC/Cy7⁻ PE⁺) were identified (Figure 5F). The proportions of Granzyme B positive (FITC⁺) cells were further calculated. The presented data of each sample was the average of three replicates.

Multiplexed immunofluorescence staining

Multiplex staining of FFPE tissue was performed using PANO 7-plex IHC kit (Cat# 0004100100, Panovue) according to manufacturer's instruction. IDO (Cat# 86630, CST), CD45 (Cat# BX00087, Biolynx), PD-L1 (Cat# 13684, CST), CD80 (Cat# MAB140, R&D Systems) and CD1C (Cat# TA505384, Origene) antibodies were sequentially applied, followed by horseradish peroxidase-conjugated secondary antibody incubation and tyramide signal amplification. The slides were microwave heat-treated after each tyramide signal amplification operation. Nuclei were stained with DAPI after all the antigens above had been labeled. The stained slides were scanned to obtain multispectral images using the Mantra System (PerkinElmer), which captures the fluorescent spectra at 20-nm wavelength intervals from 420 to 720 nm with identical exposure time. For each slide, 5 fields of immune cell enriched tumoral area were selected for image capture. The selected field were scanned to obtain multispectral images using the Mantra System, which captures the fluorescent spectra at 20-nm wavelength intervals from 420 to 720 nm with identical exposure time.

QUANTIFICATION AND STATISTICAL ANALYSIS

Single-cell and bulk RNA-seq data processing

For full-length data (Discovery cohort), the fastq data of scRNA-seq was produced using the BGISEQ500 with SE100 mode. First, the reads with the adaptor or poly-A sequences were removed from the raw fastq data before alignment using SOAPnuke (v1.5.6) (Chen et al., 2018c). The low-quality reads with "N" bases rate > 0.05 and low-quality (quality < 5) base rate accounting for 50% of the read length were also removed. Second, clean reads were aligned to the Ensembl GRCh38 using Bowtie2 (Langmead and Salzberg, 2012) with setting parameters "-q -phred33 -sensitive -dpad 0 -gbar 99999999 -mp 1,1 -np 1 -score-min L,0,-0.1." We used RSEM (RNA-Seq by Expectation Maximization) (Li and Dewey, 2011) to count the number of uniquely mapped read pairs for each gene and then gene expression was quantified by calculating the TPM (transcripts per million) value. Bulk RNA-seq data was processed using the same pipeline as scRNA-seq data.

We defined genes with TPM > 1 as detected genes. To filter out low-quality cells we set the following criterion: 1). Mapping reads ≥ 1 M; 2). Mapping rate $\geq 30\%$; 3). $1,500 \leq$ detected genes number $\leq 10,000$. In total, 16,498 cells were selected for further analysis.

For the Validation cohort 1 of 4 pairs PT and RT patients, single-cell data were generated by 10x Genomics by applying Cell Ranger (version 3.0.1, 10x Genomics) that processes Chromium single-cell RNA-seq outputs to align reads and generate the feature-barcode unique molecular identifier (UMI) matrices based on the human reference genome GRCh38.

Bulk whole exome sequencing and data processing

Low-quality sequencing reads were removed from bulk exome sequencing data using fastp (Chen et al., 2018b) with default parameters. The remained high-quality reads were aligned to the human genome build 19 (hg19) by MegaBOLT (V1.5.3.11) (Sun et al., 2020) with parameters "-type allnoindex -interval_padding 0 -stand_call_conf 30 -knownSites \$cosmic_v86.hg19.vcf.gz -vcf \$dbSNP_151.hg19.vcf.gz. This step performed alignment, sorting sequences, making duplicates, base quality score recalibration (BQSR) and SNP detection. Genome Analysis Toolkits (GATK, V3.8.1) (McKenna et al., 2010) VariantFiltration was used to apply filtering criteria to the remaining variants with parameters "-filterExpression 'QD < 2.0 || MQ < 40.0 || ReadPosRankSum < -8.0 || FS > 60.0'" for germline variants calling. a Quality of depth (QD) of 2 and Fisher strand score (FS) of 60 for SNV calling, 200 for INDEL calling. The generated Variant Call Format (VCF) files were filtered by the target region of the xGen Exome Research Panel and annotated by ANNOVAR (version Feb 2016) (Wang et al., 2010) to predict the functional consequences of the discovered variants, as well as for obtaining allele frequencies from dbSNP 151 (<https://www.ncbi.nlm.nih.gov/snp/>) and the 1000 Genomes project (<https://www.internationalgenome.org/>).

Somatic mutations were detected by comparison of each tumor to the corresponding germline DNA represented by adjacent tissue. Somatic SNVs were detected by Mutect2 (imbedded in GATK) with default parameters.

Single cell SNP & CNV calling

For single-cell SNPs calling, sorted BAM files were used to perform reads group (RG) tagging and duplicate reads marking using Picard. Then, variants were called by using GATK according to the suggested best practice following Split'N'Trim, base recalibration, variant calling, and variant filtering.

To identify malignant cells with clonal large-scale chromosomal copy number variations, we used the inferCNV R package (Patel et al., 2014) to infer the genetic profiles of each cell based on the average expression of large genes sets (101 genes) in each chromosomal region of the tumor genome compared to normal cells. All hepatic-like cells were input as interrogation group and other cells from each patient were sampled randomly as control. Other parameters were set as default.

Phylogenetic analysis from the same patient

The CCF of individual somatic variants were estimated using R package ABSOLUTE (v1.0.6) (Carter et al., 2012), which also estimates the sample purity, ploidy, and absolute allelic copy number of each mutation. Phylogenetic analysis between primary and relapsed tumor samples from the same patient, subclonal reconstruction and tree building were performed with the PhylogenicNDT package using patterns of shared mutations and CCFs with parameters “Cluster -i Patient_s -sif \$somatic_variant.sif_input.sif -ni 100-order_by_time point -rb” (Leshchiner et al., 2018).

Unsupervised clustering and markers identification

We used the Seurat R package (v2.4.3) (Satija et al., 2015) to perform unsupervised clustering of the single cells using the reads count matrix as input. First, the read counts for each cell were divided by the total counts for that cell and multiplied by the scale factor (10,000), and then natural-log transformed. To avoid unexpected noise, we only kept protein-coding genes and genes detected in less than 10 cells were excluded for downstream analysis. We performed principal component analysis (PCA) on the normalized expression matrix using highly variable genes identified by “FindVariableGenes” function. Following the results of PCA, the appropriate principal components (PCs) were selected for clustering with the specific resolution parameters (see below). Finally, to detect cluster-specific expressed genes, the clusters were compared pairwise using the Seurat “FindAllMarkers” function to test genes with > 0.25-fold difference (log-scale) on average between the two groups of cells and detectable expression in more than 50% of cells in either of the two populations.

For the clustering of all cells, the top 20 PCs were selected with a resolution parameter equal to 0.8. For the clustering of T lymphocytes, the top 12 PCs were selected with a resolution parameter equal to 0.8. For the clustering of myeloid cells, the top 16 PCs were selected with a resolution parameter equal to 0.6.

Differentially expressed gene analysis for PT and RT CD8⁺ T Cells

To define feature genes in CD8 T cells from RT, differential expression analysis between CD8⁺ T cells from PT and RT was carried out using “FindMarkers” function, implemented in the Seurat package, with log-scaled fold change ≥ 0.25 and P value < 0.05 (Wilcoxon Rank Sum test). Differentially expressed genes passing the criteria were show in Table S4.

TCR analysis

We used the TraCeR (Stubbington et al., 2016) method to assemble the TCR sequences for each single cell. TraCeR can identify the rearranged TCR chains and uses kallisto to calculate their TPM values. For every single cell, we rearranged its productive TCR chains by their TPM values. For example, if two TCR α chains were assembled in one single cell and they were both productive, the chain with the higher TPM was defined as TCR α 1 while the chain with the lower TPM as TCR α 2. Non-productive TCR chains were excluded. The same rearrangement analysis was deployed on TCR β . We kept cells with at least one productive TCR α or TCR β chain for subsequent analysis. For cells with only one TCR α and one β chain, cells with identical TCR α or β chains were defined as a T cell clone. Meanwhile, for cells with paired TCR α and β chains, cells shared at least one pair of identical TCR α and β chains were defined as one clone, while identical paired TCR α and β chains which were shared by three or more cells were defined as an expanded clone.

The droplet-based TCR-seq data of P34 was processed using Cell Ranger (version 3.0.1, 10x Genomics) against the human GRCh38 vdj reference provided by 10x Genomics. We identified T cell clonotype shared by both PT and RT CD8⁺ T cells with the same TCR α and β CDR3 sequences.

The bulk TCR sequencing data were analyzed using TCR and BCR repertoire analyzing pipeline iMonitor (Zhang et al., 2015). First, basic quality control is performed to discard low-quality reads and clean paired-end reads were merged by their overlapping nucleotides. Second, the merge reads were assigned to VDJ germline segments and alleles from IMGT (<http://www.imgt.org/>) database by BLAST alignment and further re-alignment. Third, sequencing error was corrected. Then we compared clonotypes between adjacent normal tissue and tumor by calculating the top 10 abundant clonotypes sharing the same CDR3 of normal or tumor tissues.

Cell developmental trajectory

The cell lineage trajectory of CD8⁺ T was inferred by using Monocle2 (Qiu et al., 2017). We excluded MAIT cells according to their TCR identity due to their distinct TCRs and development processes relative to other CD8⁺ cells. We first used the “relative2abs” function in Monocle2 to convert TPM into normalized mRNA counts and created an object with parameter “expressionFamily = neg-binomial.size” following the Monocle2 tutorial. We used the “differentialGeneTest” function to derive DEG from each cluster and

genes with a q -value $< 1e-5$ were used to order the cells in pseudotime analysis. After the cell trajectories were constructed, differentially expressed genes along the pseudotime were detected using the “differentialGeneTest” function.

Definition of cell scores and signature

We used the average expression (measured by $\log_2(\text{TPM} + 1)$) of 5 resident markers (*RUNX3*, *NR4A1*, *CD69*, *CXCR6*, and *NR4A3*), 7 cytotoxicity associated genes (*PRF1*, *IFNG*, *GNLY*, *NKG7*, *GZMB*, *GZMA*, *CST7*, and *TNFSF10*), 5 exhausted markers (*CTLA4*, *HAVCR2*, *LAG3*, *PDCD1*, and *TIGIT*) and 6 costimulatory molecular genes (*ICOS*, *CD226*, *TNFRSF14*, *TNFRSF25*, *TNFRSF9*, and *CD28*) to define the resident, cytotoxic, exhausted, and costimulatory score for CD8 T cells.

To assign M1/M2 polarization and pro-/anti-inflammatory ability estimates to macrophage cells, we applied Gene Set Variation Analysis (GSVA) (Hänzelmann et al., 2013) using standard settings, as implemented in the GSVA package. The gene sets associated with the above functions were described by Azizi et al. (2018) (Table S3).

To define metabolism phenotypes in CD8⁺ T cells in PT and RT, the metabolism signature determined as the mean expression of gene signatures involved in glycolysis, TCA cycle, lipolysis, lipogenesis, and glutamine metabolism obtained from PathCards (<https://pathcards.genecards.org/>).

The features of malignant cells were defined with the mean $\log_2(\text{TPM} + 1)$ normalized expression of hepatic, proliferation, immune surveillance, or immune escape genes (Table S3).

Pathway analysis

To investigate biological states or function differences of cells from primary/relapsed tumor, the DEGs between primary/relapsed malignant cells were used to investigate Hallmark gene sets by Gene Set Enrichment Analysis (GSEA) (Subramanian et al., 2005; Mootha et al., 2003). The phase gene sets of CD8 T cell trajectory were enriched with the Hallmark gene sets and Reactome pathway by GSEA. Pathway enrichment analysis for DEG between CD8 cells from PT and RT, Ingenuity Pathway Analysis (IPA) was performed to characterize the biological functions of cells. The differentially expressed genes between different cell types were uploaded into the IPA software for the core analysis, and identified the canonical pathways, diseases and functions, upstream regulators, and gene networks.

Survival analysis

For IHC samples, we analyzed the association of CD8⁺CD161⁺ T cell number and second relapse in RT cohort. The samples were grouped into high and low groups based on the optimal cutpoint determined by R function `surv_cutpoint`. Kaplan–Meier survival curves with the cumulative number of events table and the cumulative number of censored subjects table were plotted by `survminer` package.

In the bulk Validation cohort and The Cancer Genome Atlas (TCGA) Liver Cancer (LIHC) data, the RT *KLBR1*^{hi} T cell signature was defined as the mean $\log_2(\text{TPM} + 1)$ normalized expression of feature genes and divided by the expression of *CD8A*. The *KLBR1*^{hi} CD8 T cell feature genes included *KLBR1*, *CCL20*, *CCR6*, *IFNGR1*, *IL7R*, *EGR1*, *SATB1*, *SLC4A10*, *NFKB1A*, *GPR183*, *LDHA*, *VIM*, *RORA*, *CREM*, *NINJ1*, *NCR3*, *TWISTNB*, *REL*, *TTC39C*, *CA2*, *G3BP2*, *NFKB1*, *EIF4A3*, *FTH1*, *GPR65*, *SKIL*, *THBS1*, *CD83*, *SQSTM1*, *TRMO*, *GNG2*, *JCHAIN*, *ADAM19*, *ATP1B3*, *PDCD4*, *ANXA1*, *BEST1*, *FOS*, *STAT4*, *IL4I1*, *ODF2L*, *FKBP11*, *TC2N*, *TNFAIP3*, *KDM6B*, *INSIG1*, *SPOCK2*, *GLIPR1*, *BIRC3* and *YPEL5*. For the bulk validation cohort 2, the survival analysis was performed using `survminer` package. The TCGA LIHC gene expression and survival data were downloaded from the University of California, Santa Cruz (UCSC) Xena (<http://xena.ucsc.edu/>). The samples were grouped into high and low expression groups by the median value. Kaplan–Meier survival curves were plotted to show differences in survival time, and log-rank p values reported by the Cox regression models implemented in the R package `survival` were used to determine the statistical significance.

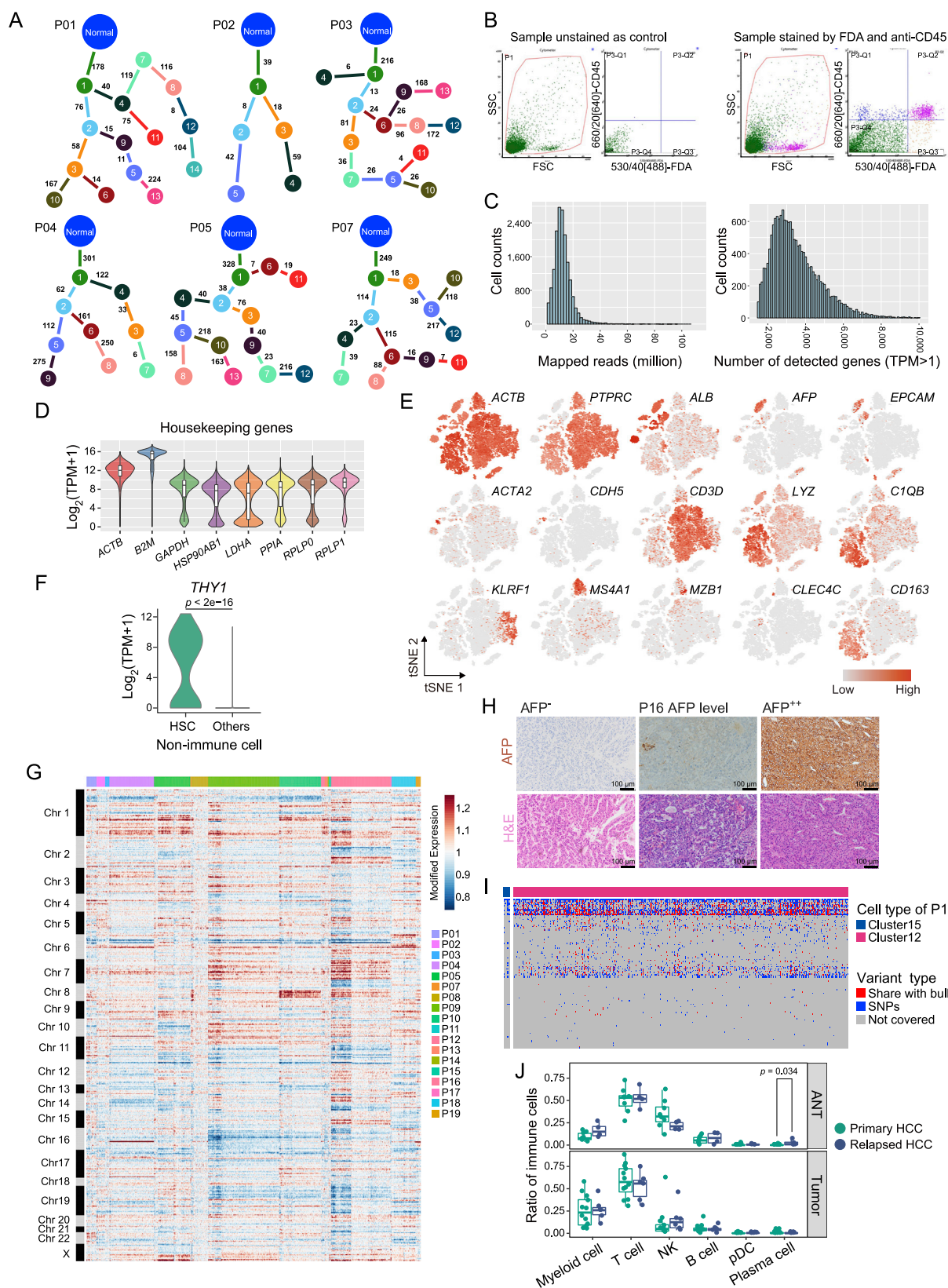
Cell-cell interaction analysis

To analyze cell-cell interactions between different cell types, we used CellPhoneDB (Vento-Tormo et al., 2018) to identify significant ligand-receptor pairs within primary and relapsed HCC samples. For both primary and relapsed HCC, the cell type-specific receptor-ligand interactions between cell types were identified based on the specific expression of a receptor by one cell type and a ligand by another cell type. The interaction score refers to the total mean of the individual ligand-receptor partner average expression values in the corresponding interacting pairs of cell types. The expression of any complexes output by CellPhoneDB was calculated as the sum of the expression of the component genes.

Statistics

Cell distribution comparisons between two groups were performed using unpaired two-tailed Wilcoxon rank-sum tests. Comparisons of gene expression or gene signature between two groups of cells were performed using unpaired two-tailed Student's t test. Comparisons of cell distribution of paired RT and PT as well as tumor and ANT were performed using paired two-tailed Wilcoxon rank-sum tests. All statistical analyses and presentation were performed using R. Statistical tests used in figures were shown in figure legends and statistical significance was set at $p < 0.05$. Exact value of n was shown in the figures and figure legends and what n represents was shown in the figure legends.

Supplemental Figures



(legend on next page)

Figure S1. ScRNA-seq profiling of the ecosystem in primary and relapsed HCC samples, related to Figure 1

- (A) Phylogenetic analysis, based on somatic mutations. Shared somatic events from primary and recurrent tumors are shown in hierarchical clusters. The numbers of each cluster are provided in each circle. The blue circles on the top of the panel represent the starting points of somatic mutations.
- (B) Two-dimensional scatterplot, showing the gating strategy used to sort live CD45⁺ immune cells (P3-Q2 in the right panel) and CD45⁺ non-immune cells (P3-Q3 in the right panel). Sorted cells were used as the template for single-cell RNA preparations.
- (C) Histogram indicating the total number of mapped clean reads (left panel) and the number of detected genes (TPM > 1) (right panel).
- (D) Violin plot showing the expression levels of 8 known housekeeping genes, in all cells.
- (E) tSNE plot showing the expression levels of marker genes, defined for all cell types.
- (F) Violin plots showing *THY1* expression in HSCs and other non-immune cells, including HCC malignant cells and endothelial cells. The p value shown on the top of the plot was calculated using Student's t test.
- (G) Heatmap representing the CNV analysis, inferred from the single-cell RNA-seq data for each tumor cell. Each column represents a single cell and each patient analyzed is color-coded.
- (H) IHC staining showing AFP expression levels or hematoxylin-eosin staining in AFP⁺ (left), P16 (middle), and AFP⁺ (right) cells. The scale bars represent 100μm.
- (I) Colored plot showing the comparisons in the detected mutations between single-cell analyses and matched bulk tumor analyses. Each row shows the mutations detected in the bulk WES of the tumor, relative to the adjacent non-tumor tissue control. Each column indicates a single cell. Red and blue colors indicate cells showing the same or different variants compared to the bulk tumor, respectively. Grey blocks indicate no reads covering the mutation region.
- (J) Boxplot showing the fractions of main cell types in total immune cells derived from primary and relapsed patients in adjacent non-tumor tissue (ANT) and HCC tumor tissue, respectively. Significance was calculated using the Wilcoxon test.

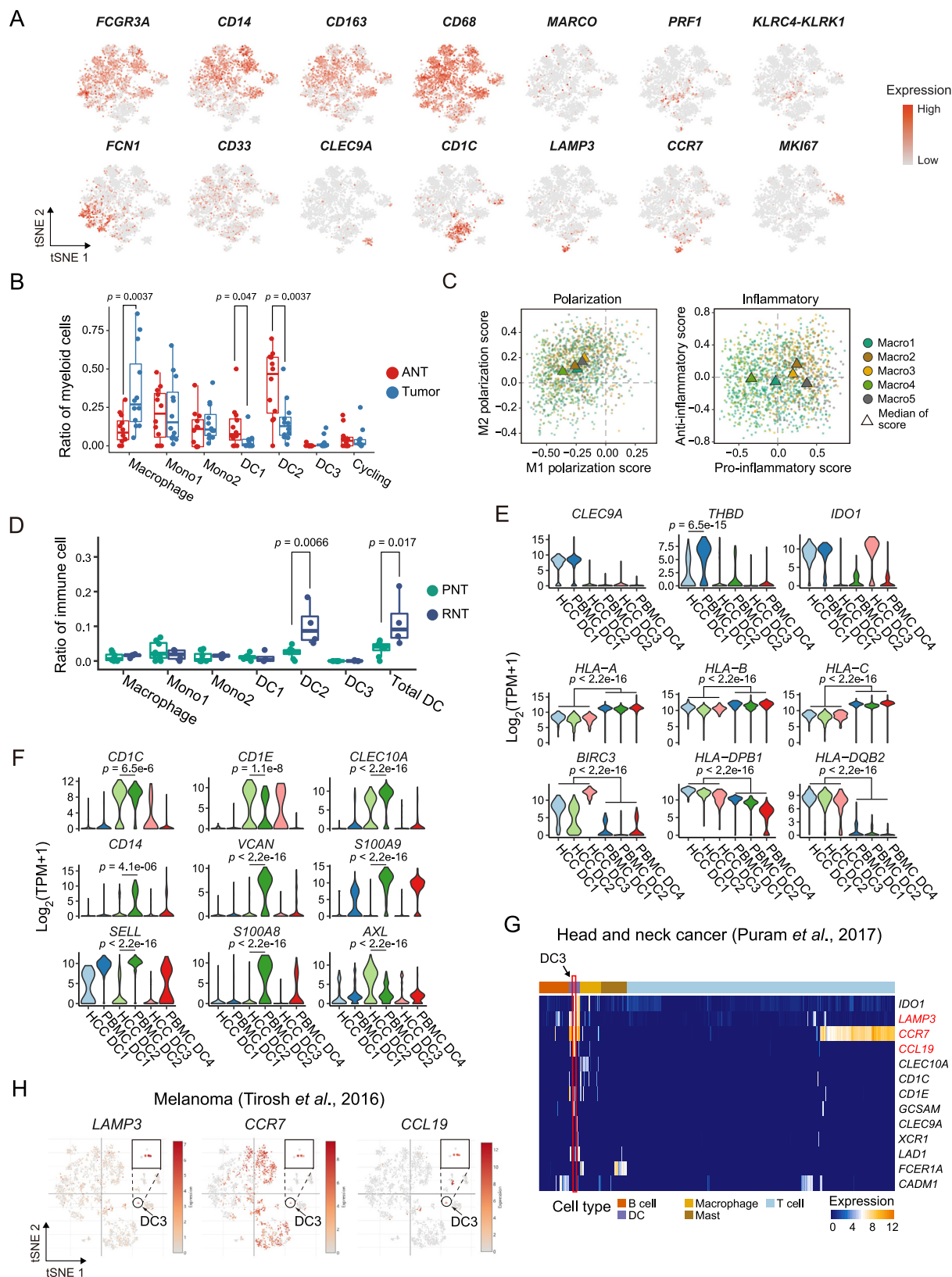


Figure S2. Myeloid cell subtypes in HCC, related to Figure 2

(A) tSNE visualized plot showing the expression of selected marker genes for the defined cell types.

(B) Boxplot showing the fraction of myeloid subgroups in tumor tissue (blue) and ANT (red). The p value indicated in the plot was calculated by a paired Wilcoxon test.

(C) Scatterplots showing M1 and M2 scores (left panel) and pro- and anti-inflammatory scores (right panel) for each color-coded macrophage subtype.

(D) Boxplot showing the fraction of myeloid subgroups in PNT (green) and RNT (blue). Significance was determined by the Wilcoxon test.

(E) Violin plots showing the expression of selected genes in DC1 cells from HCC and PBMC samples.

(F) Violin plots indicating the expression of selected genes in DC2 cells from HCC and PBMC samples. Significance for the E and F panels was calculated by Student's t test.

(G) Log₂ scaled expression of DC subtype markers (rows) in publicly available head and neck cancer single-cell datasets.

(H) tSNE plot showing the expression of DC3 markers (*CCR7*, *LAMP3*, and *CCL19*) in published melanoma data.

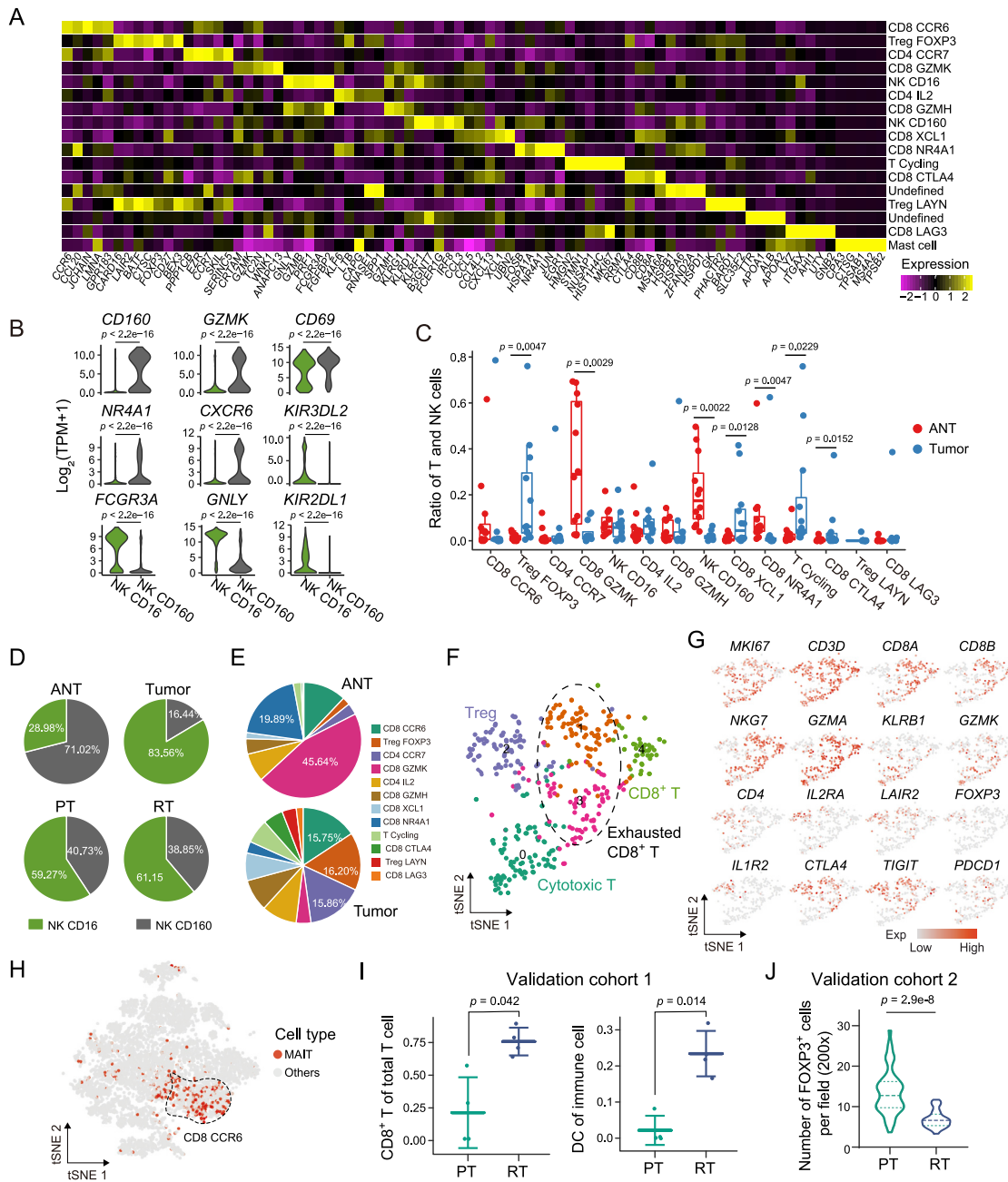


Figure S3. The subtypes of T/NK cells in HCC, related to Figure 3

(A) Heatmap showing the scaled, average expression of typical marker genes for each of the clusters defined in Figure 3A.

(B) Violin plot showing the expression of marker genes in NK CD160 and NK CD16 cells. P value was calculated by Student's t test.

(C) The boxplot shows the fraction of each subtype of T/NK cell clusters in ANT (red) or tumor tissues (blue). Significance was determined by paired Wilcoxon test.

(D) Pie chart showing the distribution of NK CD160 and NK CD16 cells in ANT, tumor, PT, and RT.

(E) Pie chart showing the distribution of T cell subtypes in ANT and tumor tissues.

(F) tSNE visualization of re-clustered cycling T cells. The dashed circle indicates the exhausted CD8⁺ T cell cluster.

(G) tSNE plot showing the expression of marker genes for the identification of the indicated cell subtypes in panel F.

(H) tSNE projection of T cells and NK cells, with red color-coding representing the specific TCR combination of MAIT (mucosal-associated invariant T) cells.

(I) Bar plot showing the fraction of CD8⁺ T cells and DCs, based on scRNA-seq data from 4 additional pairs of primary (PT, green) and relapsed tumors (RT, blue) from the validation cohort 1. Significance was calculated by paired Student's t test.

(J) Violin plot showing the number of FOXP3⁺ cells (Tregs) in paired PT (green) and RT (blue) samples based on IHC result from Validation cohort 2. P values were calculated by paired Wilcoxon test.

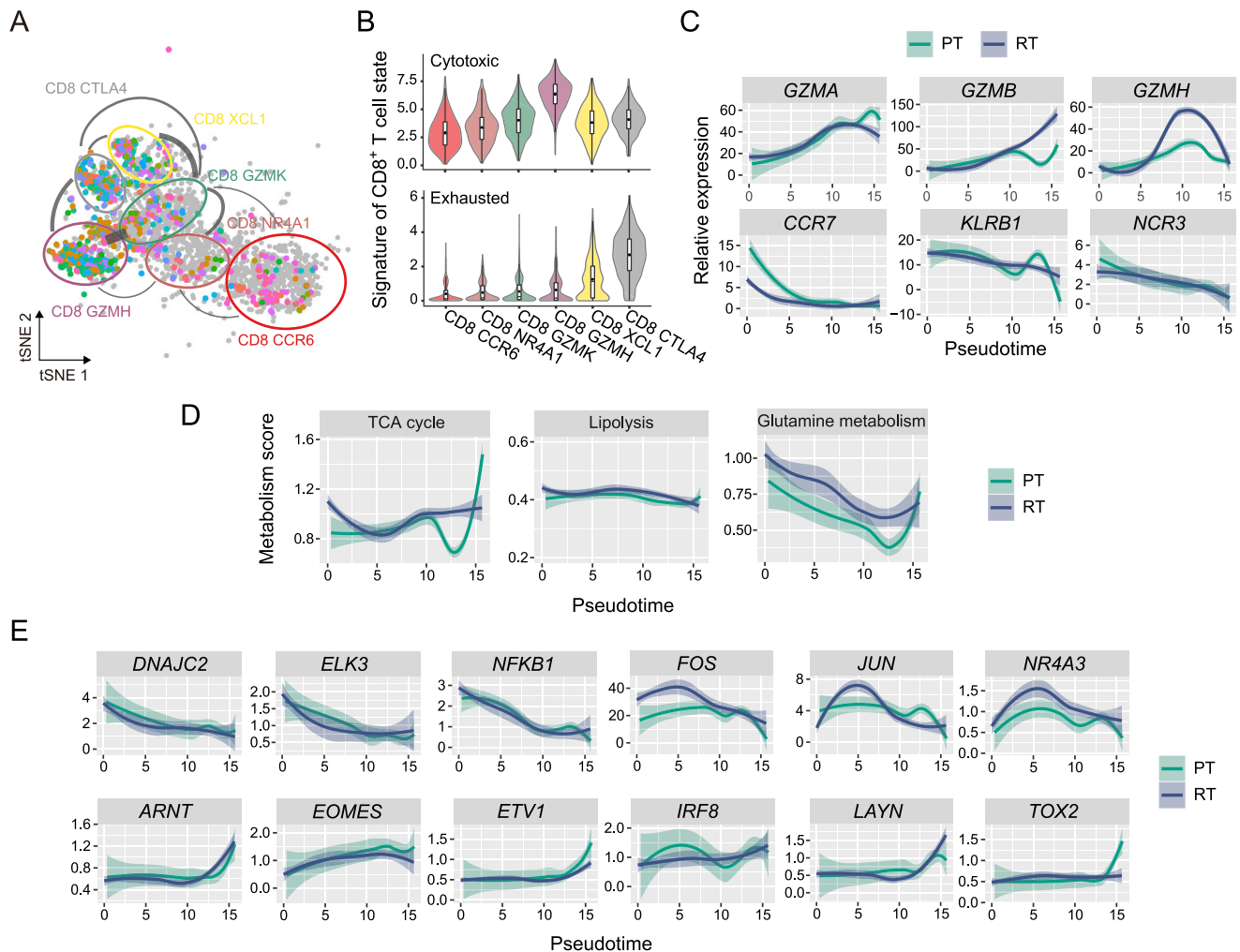


Figure S4. Analysis of CD8⁺ T cell transition states in RT and PT samples, related to Figure 4

(A) tSNE plot showing common clones shared among the different color-coded T cell subtypes. The thickness of the connecting lines between subtypes represents the number of shared TCR clones.

(B) Violin plot indicating the cytotoxic (top) or exhausted (bottom) gene expression signatures in T cell subtypes.

(C) Shaded line plot indicating the expression levels of the selected genes along the pseudotime in PT and RT samples.

(D) Shaded line plot showing the expression scores for the TCA cycle, lipolysis, and glutamine metabolism pathways along the pseudotime in PT (green) and RT (blue) samples.

(E) Shaded line plot indicating the expression levels of the selected transcription factors and genes along the pseudotime in PT and RT samples.

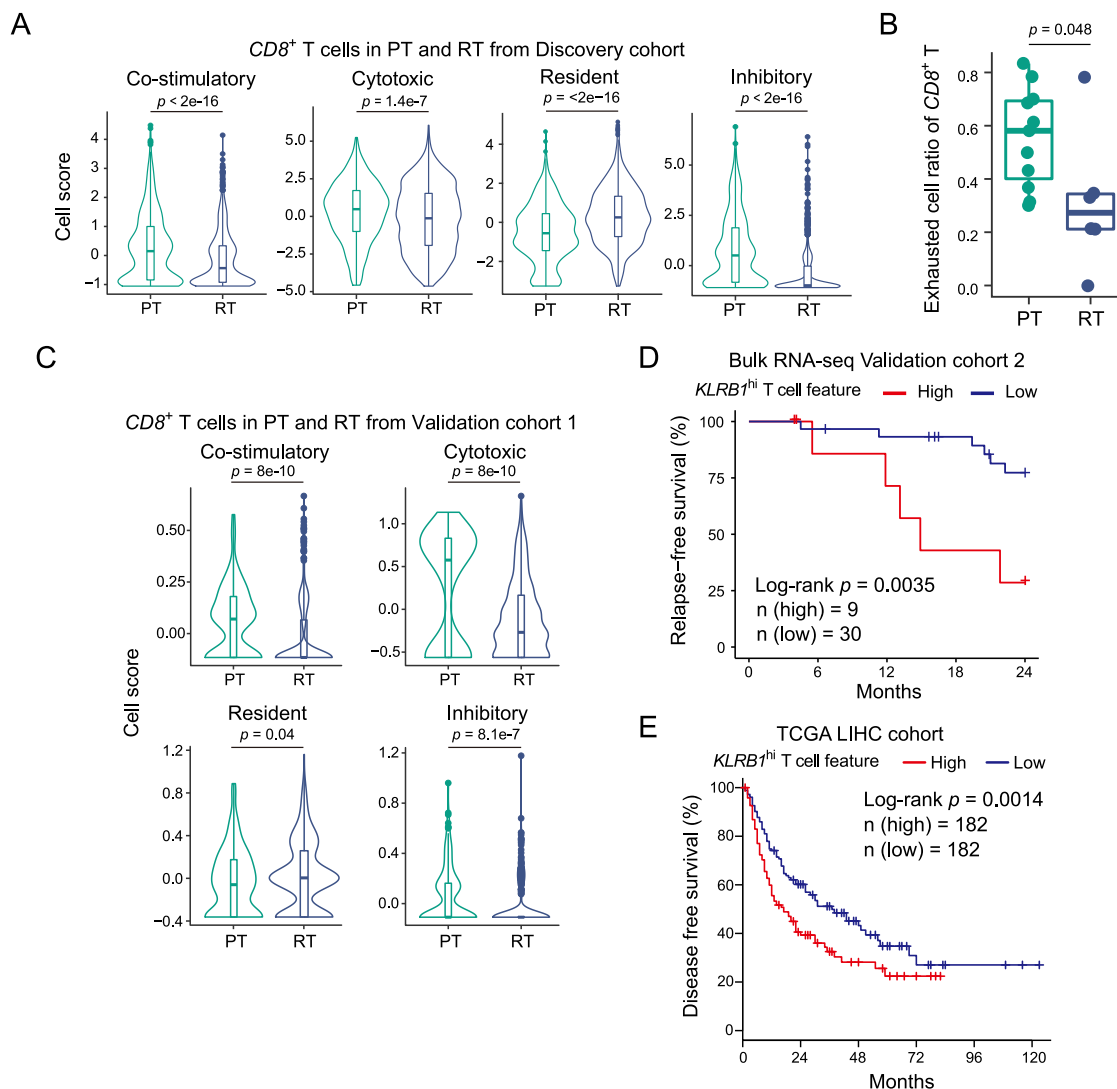


Figure S5. *CD8⁺* T cells detected in the innate-like state in RT samples, related to Figure 5

(A) Violin plot showing the expression of co-stimulatory, cytotoxic, resident, and inhibitory signature genes in *CD8⁺* T cells in PT (green) and RT (blue) samples from the Discovery cohort. Significance was determined by Student's *t* test.

(B) Boxplot indicating the proportion of *CD8⁺* T cells characterized by the high expression of exhausted signature genes. P values were determined by the Wilcoxon test.

(C) Violin plot indicating the expression of co-stimulatory, cytotoxic, resident, and inhibitory signature genes in *CD8⁺* T cells in PT (green) and RT (blue) samples from the Validation cohort 1. Significance was determined by Student's *t* test.

(D-E) Kaplan-Meier plot of disease-free survival in the Validation cohort 2 (D) and the publicly available TCGA LIHC cohort (E), comparing low versus high *KLRB1^{hi}**CD8⁺* T cell patients. The numbers of patients from each subgroup and the Log-Rank *t* test results are indicated in the plot. N represents the number of patients.

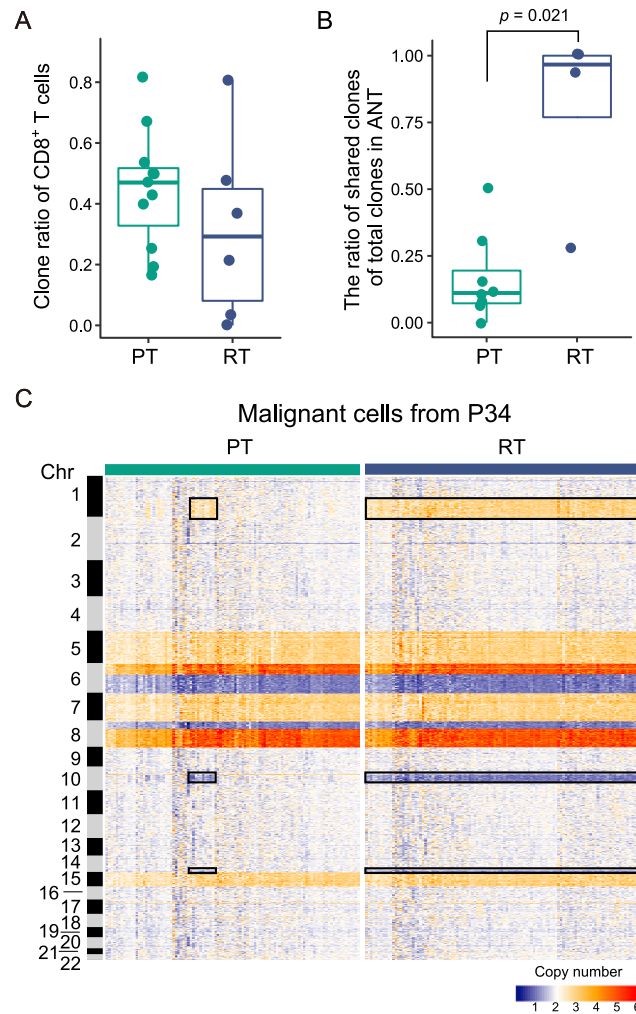


Figure S6. CD8⁺ T cells showed reduced clonal expansion in relapsed tumors, related to Figure 6

(A) Boxplot indicating the fraction of clonal T cells from total CD8 T cell populations in PT (green) and RT (blue) samples.

(B) Boxplot showing the fraction of shared T cell clones between ANT and tumors, in primary HCC and relapsed HCC, respectively. The p values shown in the plot were determined by the Wilcoxon test.

(C) Heatmap representing the CNV profiling of malignant cells in paired RT and PT samples from patient P34.

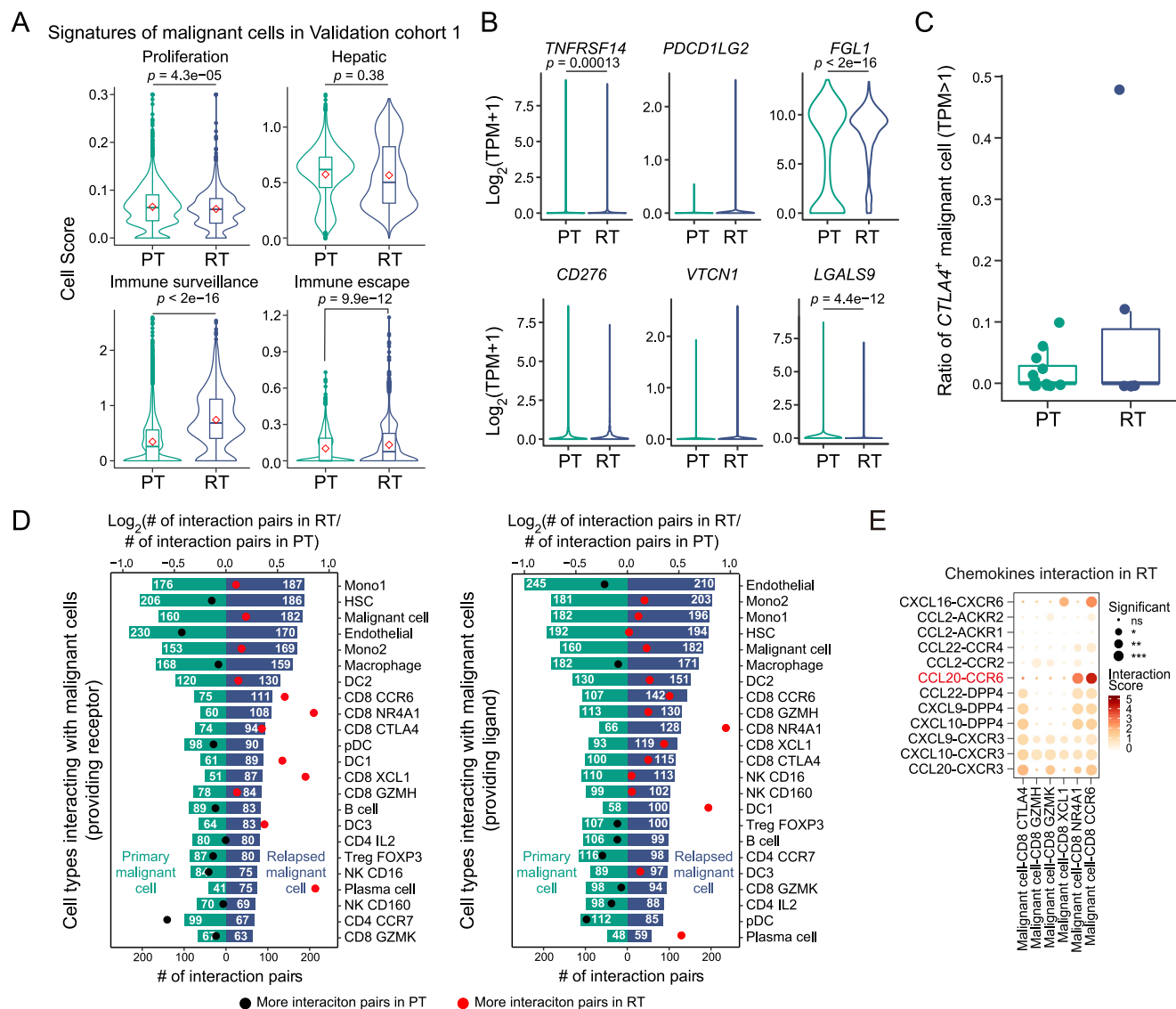


Figure S7. The contributions of relapsed malignant cells to the tumor ecosystem, related to Figure 7

(A) Violin plots showing proliferation, hepatic scores, immune escape, and immune surveillance scores of malignant cells from PT (green) and RT (blue) patients in the Validation Cohort 1.

(B) Violin plot representing ligand/receptor expression levels for the immune checkpoint genes in primary (green) and relapsed (blue) malignant cells. P values shown in the plot were determined by Student's t test.

(C) Boxplot indicating the fraction of CTLA4⁺ tumor cells in PT (green) and RT (blue) samples.

(D) Bar chart showing the number of significant ligand-receptor pairs in malignant cells and other cells in PT and RT samples. Malignant cells providing ligands (left panel) and receptors (right panel) were calculated separately. The dots represent the ratio of the number of significant Ligand-Receptor pairs between RT and PT samples, the ratio above 1 in red and below 1 in black.

(E) Bubble chart showing the interactions between chemokines (released by malignant cells) and their receptors (expressed by CD8 T cell subtypes) in RT samples. The scores represent the normalized expression levels, and the sizes of the bubbles indicate the significance of the interaction, as calculated by CellPhoneDB (STAR methods).